

**Ceylon Rail : AI-Driven Intelligent Railway Operations
System for Sri Lanka**

INDIVIDUAL THESIS

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Individual component: Advanced Intelligent Ticket Fraud Detection and
Risk-Aware Passenger Verification

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DECLARATION AND SUPERVISOR CERTIFICATION

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ABSTRACT

Railway ticket fraud and invalid ticket usage can negatively affect railway revenue, passenger management, and operational efficiency. Traditional ticket inspection and rule-based validation methods may be limited in identifying complex or unusual transaction patterns, particularly when large volumes of railway transaction data are involved. This research focuses on developing an intelligent railway fraud detection and risk-aware passenger verification component for the CeylonRail Intelligence platform, with particular consideration of the Sri Lankan railway context.

The study uses a quantitative and experimental research approach based on a railway transaction dataset containing 9,800 records and 18 attributes covering the period from 2020 to 2024. The dataset includes transaction, passenger, route, fare, payment, booking, ticket scanning, inspection, and fraud-related information. A time-based evaluation approach is used, with 2,750 transactions from 2024 used as the test set. Identifying information such as passenger names, passenger IDs, and transaction IDs is excluded from predictive modelling.

Three machine-learning-based anomaly detection approaches, namely Isolation Forest, One-Class Support Vector Machine (One-Class SVM), and Autoencoder, are implemented and evaluated. The models are assessed using precision, recall, F1-score, ROC-AUC, and false-positive rate. In addition, a risk-scoring mechanism is used to categorize transactions into low-, medium-, and high-risk groups, while rule-based validation is incorporated to support passenger and ticket verification.

The experimental results demonstrate the applicability of machine-learning-based anomaly detection to railway transaction fraud detection. The evaluated models identify suspicious transaction patterns with different levels of detection performance, while the risk-scoring mechanism provides an interpretable way to prioritize potentially fraudulent transactions for further inspection. The results also demonstrate the importance of balancing fraud detection capability with false-positive reduction in practical railway environments.

The developed component is implemented as part of the wider CeylonRail Intelligence platform using a Python-based machine-learning environment, Flask backend, React frontend, and MySQL database. The research demonstrates the technical feasibility of combining anomaly detection, risk scoring, and railway-specific validation within an intelligent railway management platform. The findings provide a foundation for future improvements using larger datasets, improved fraud labels, real-time transaction processing, and enhanced passenger verification mechanisms.

Keywords: Railway Fraud Detection, Anomaly Detection, Machine Learning, Isolation Forest, One-Class SVM, Autoencoder, Risk Scoring, Passenger Verification, Railway Ticketing, CeylonRail Intelligence.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
AUC	Area Under the Curve
CPU	Central Processing Unit
CSV	Comma-Separated Values
F1	F1-Score
FPR	False Positive Rate
HTTP	Hypertext Transfer Protocol
IF	Isolation Forest
LOF	Local Outlier Factor
ML	Machine Learning
MySQL	Mongo Database
QR	Quick Response
ROC	Receiver Operating Characteristic
SVM	Support Vector Machine
UI	User Interface
URL	Uniform Resource Locator

1 INTRODUCTION

1.1 Overall Project Context

Railway transportation provides affordable and accessible mobility while supporting economic activity, social connectivity, and regional movement. Efficient railway operations depend on several interconnected factors, including passenger demand, train capacity, scheduling, service reliability, ticketing, revenue management, and infrastructure utilization. The increasing availability of digital railway data creates opportunities to apply Artificial Intelligence (AI), machine learning, data analytics, and optimization techniques to improve railway decision-making.

The CeylonRail research project is developed as an AI-driven railway management system designed for the Sri Lankan railway environment. Traditional railway operations often rely on historical information, manually defined rules, and periodic monitoring, which can become challenging when dealing with large transaction volumes, changing passenger demand, delays, congestion, and operational uncertainty.

The proposed CeylonRail platform integrates several intelligent functions through a centralized architecture consisting of user access, core intelligence, shared railway knowledge, data services, and database layers. These components can exchange relevant information to support coordinated railway decision-making. For example, demand predictions can support capacity planning, delay predictions can assist scheduling, and fraud assessments can support ticket verification.

The overall project therefore aims to improve railway management, resource utilization, passenger services, and operational efficiency through an integrated intelligent decision-support environment. This dissertation focuses specifically on the AI-Based Railway Fraud Detection and Passenger Verification Module.

1.2 Individual Component Context

The individual component focuses on identifying potentially fraudulent railway ticketing activities, assessing suspicious passenger and ticket behavior, and supporting verification of legitimate passenger-ticket combinations.

Digital ticketing generates valuable transactional and behavioral information that can be analyzed for automated validation and fraud detection. However, traditional ticket inspection relies heavily on manual verification and may have difficulty identifying suspicious patterns distributed across multiple transactions.

The proposed component considers information such as ticket transactions, passenger travel information, route usage, purchase frequency, cancellation behavior, ticket scanning behavior, inspection status, and fare-related information. These data are transformed into meaningful features representing patterns such as duplicate ticket usage, abnormal booking frequency, excessive cancellations, route inconsistencies, and unusual travel behavior.

The proposed approach applies machine-learning-based anomaly detection using **Isolation Forest, One-Class SVM, and Autoencoder** models, supported by a rule-based railway validation mechanism. The machine-learning models identify transactions that differ from learned patterns of normal behavior, while rule-based validation supports explicit railway-specific checks. The component therefore addresses two related functions: intelligent fraud detection and passenger-ticket verification.

The module operates within the wider CeylonRail architecture, allowing fraud-related outputs to contribute to the overall railway intelligence environment..

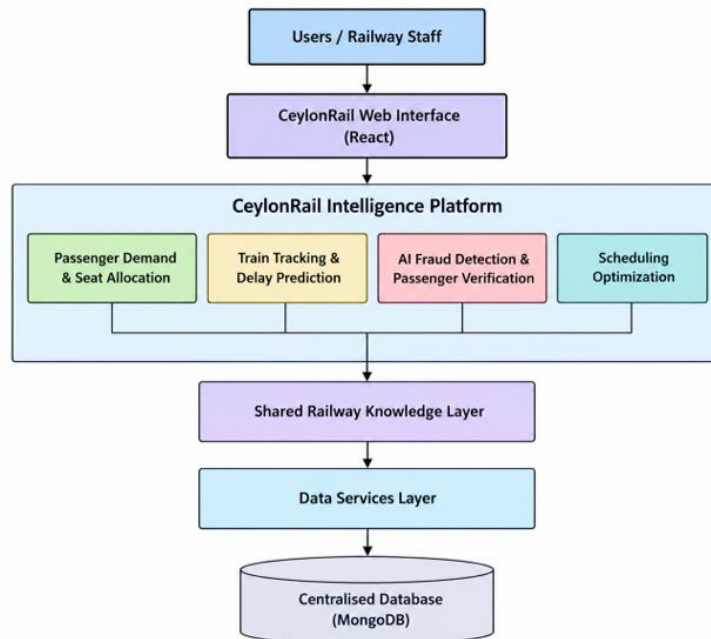


Figure 1 - Overall System Diagram

1.3 Problem Statement

Railway ticket fraud and invalid ticket usage can create revenue and operational challenges. Conventional verification mainly relies on manual inspection and predefined rules, which may identify obvious inconsistencies but have limited ability to detect complex behavioral patterns across historical transactions.

Suspicious activities such as abnormal booking frequency, repeated cancellations, duplicate ticket usage, route inconsistencies, and unusual travel behavior may require analysis of multiple transactions rather than a single ticket. Furthermore, manually reviewing large numbers of transactions can be time-consuming and may prevent railway personnel from efficiently prioritizing suspicious cases.

Machine learning can assist in identifying behavioral patterns, but it cannot independently represent all railway-specific requirements related to ticket validity, ownership, routes, and passenger information. Therefore, both intelligent fraud detection and explicit validation mechanisms are required.

The research problem is therefore defined as follows:

Railway ticket verification processes that primarily rely on manual and conventional validation may have limited capability to identify complex fraudulent and abnormal ticketing behavior. Therefore, an intelligent approach is required that combines transactional and behavioral analysis, machine-learning-based anomaly detection, risk assessment, and railway-specific validation rules to detect suspicious activities and support reliable passenger-ticket verification.

1.4 Research Gap

Existing railway research has applied AI and machine learning to areas such as passenger forecasting, delay prediction, ticket verification, fraud detection, and scheduling. Research has also explored blockchain-based ticketing, smart validation, facial recognition, behavioral fraud detection, and machine-learning-based approaches. However, several gaps remain relevant to an integrated railway fraud detection solution.

1.4.1 Limited Integration of Fraud Detection and Railway Validation

Machine-learning methods can identify behavioral patterns, while rule-based methods can enforce explicit railway requirements. Treating these approaches separately may limit their effectiveness. An integrated approach can combine the flexibility of machine learning with railway-specific validation rules.

1.4.2 Limited Consideration of Multiple Behavioral Indicators

Fraudulent behavior may involve several indicators, including booking frequency, cancellation behavior, route usage, duplicate tickets, scanning behavior, and passenger travel patterns. Considering multiple indicators can provide a more comprehensive representation of suspicious behavior.

1.4.3 Need for Risk-Oriented Fraud Detection

Binary fraud classification may not provide sufficient information for operational decision-making. A fraud risk score can help identify the severity of suspicious activity and prioritize cases for further investigation.

1.4.4 Separation of Fraud Detection and Passenger Verification

Fraud detection identifies suspicious patterns, while passenger verification determines whether the passenger and ticket satisfy required conditions. Integrating these functions can provide more comprehensive ticket security.

1.4.5 Sri Lankan Railway Context

There is also a need for solutions specifically aligned with the Sri Lankan railway environment. The proposed component considers local railway routes, ticketing requirements, passenger information, and verification processes within the CeylonRail system.

1.4.6 Overall Research Gap

The research therefore addresses the need for an integrated approach combining transactional and behavioral anomaly detection, railway-specific validation, risk assessment, and passenger-ticket verification within an intelligent railway management environment.

1.5 Aim and Objectives

1.5.1 Research Aim

The primary aim of this research is:

To develop an AI-based railway fraud detection and passenger verification component that identifies suspicious ticketing behavior, generates fraud risk assessments, and supports reliable verification of railway tickets and passengers within the CeylonRail intelligent railway management system.

1.5.2 Research Objectives

1. **Identify railway ticket fraud patterns and relevant indicators** such as duplicate usage, abnormal booking frequency, cancellations, route inconsistencies, scanning behavior, and unusual travel behavior.
2. **Develop transactional and behavioral features** from available ticketing, passenger, route, fare, payment, booking, scanning, inspection, and temporal information.
3. **Develop and evaluate machine-learning-based anomaly detection models** using Isolation Forest, One-Class SVM, and Autoencoder approaches to identify potentially suspicious railway transactions.
4. **Integrate machine learning with railway-specific validation rules** to combine behavioral anomaly detection with explicit railway requirements.
5. **Develop fraud risk scoring and passenger-ticket verification capabilities** to support the identification and prioritization of transactions requiring further investigation.
6. **Evaluate the proposed component** using appropriate performance measures including precision, recall, F1-score, ROC-AUC, and false-positive rate.

1.6 Research Questions

RQ1:What transactional and behavioral characteristics can be used to identify potentially fraudulent railway ticketing activities?

RQ2:How effectively can machine-learning-based anomaly detection identify potentially fraudulent railway ticketing activities?

RQ3:How can railway-specific validation rules be integrated with machine-learning-based fraud detection?

RQ4:How can fraud risk scoring support railway ticket verification and operator decision-making?

RQ5:How effectively can the proposed component support the verification of passenger-ticket combinations?

RQ6:What is the performance of the proposed anomaly-detection approaches in terms of precision, recall, F1-score, ROC-AUC, and false-positive rate?

1.7 Scope, Limitations and Assumptions

1.7.1 Scope of Research

The research focuses on the development and evaluation of the AI-Based Railway Fraud Detection and Passenger Verification Module within CeylonRail.

The component analyses railway transaction and passenger-related information to identify potentially fraudulent or abnormal behavior. The main indicators include:

- Duplicate ticket usage
- Abnormal booking frequency
- Excessive cancellations
- Route inconsistencies
- Unusual travel behavior
- Ticket scanning behavior
- Fare and transaction inconsistencies
- Inspection-related information

The research includes feature engineering, machine-learning-based anomaly detection using Isolation Forest, One-Class SVM, and Autoencoder models, fraud risk scoring, and rule-based ticket and passenger verification.

The component is designed to support railway operators rather than completely replace human decision-making. It is also integrated with the wider CeylonRail architecture where applicable.

1.7.2 Limitations of the Research

The main limitations include:

- **Data availability:** Limited or unrepresentative data may affect model performance.

- **Fraud diversity:** New or previously unseen fraud patterns may not be detected effectively.
- **Data quality:** Missing, inaccurate, or inconsistent information may affect detection and verification.
- **False positives and negatives:** Legitimate transactions may be flagged, while some fraudulent activities may remain undetected.
- **Operational validation:** Research or simulated evaluation may not represent full-scale railway deployment.
- **Verification dependency:** The effectiveness of passenger and ticket verification depends on the availability and reliability of the relevant ticket, passenger, route, scanning, and inspection information.

1.8 Individual Contribution

The overall CeylonRail project is a group-based research project containing several intelligent railway components. The individual contribution of this dissertation is the design, development, and evaluation of the AI-Based Railway Fraud Detection and Passenger Verification Module.

1.8.1 Research and Problem Analysis

The research investigates railway ticket fraud and passenger verification requirements, focusing on suspicious behaviors such as duplicate ticket usage, abnormal booking frequency, cancellations, route inconsistencies, and unusual travel patterns.

1.8.2 Feature Engineering

Transactional and behavioral information is transformed into relevant fraud detection features, including:

- Ticket transaction information
- Passenger-related information
- Route usage
- Purchase and booking behavior
- Cancellation behavior
- Ticket scanning behavior
- Inspection status
- Fare-related information
- Temporal characteristics

1.8.3 Fraud Detection Models

Three machine-learning-based anomaly detection approaches are implemented:

- **Isolation Forest** – identifies observations that can be isolated relatively easily from the normal transaction population.
- **One-Class SVM** – learns a boundary representing normal transaction behavior and identifies observations outside the learned boundary.
- **Autoencoder** – learns to reconstruct transaction features and uses reconstruction error to identify potentially abnormal transactions.

The three approaches are experimentally evaluated to determine their effectiveness in detecting potentially fraudulent railway transactions.

1.8.4 Rule-Based Railway Validation

A rule-based validation mechanism is integrated with the machine-learning component to perform explicit railway-specific checks and complement anomaly-detection results.

1.8.5 Passenger and Ticket Verification

The component supports passenger-ticket verification using available ticket, passenger, route, fare, scanning, and inspection information. These verification results are considered together with the machine-learning anomaly outputs and risk assessment.

1.8.6 Risk Assessment and Alerts

The module generates fraud risk scores to help identify transactions requiring additional attention. Risk categories and verification results can assist railway operators in prioritizing suspicious activities.

1.8.7 Testing and Evaluation

The proposed approaches are evaluated using quantitative performance measures including precision, recall, F1-score, ROC-AUC, and false-positive rate. The evaluation compares the performance of Isolation Forest, One-Class SVM, and Autoencoder using the defined test data.

1.8.8 Integration with the Overall Project

The fraud detection and passenger verification module is designed to operate within the wider CeylonRail Intelligence architecture. Its services and outputs are integrated with the overall platform where applicable, allowing the individual component to contribute to the project's broader intelligent railway management objectives.

1.9 Thesis Structure

This dissertation is organized into chapters that present the research from its background and motivation through methodology, implementation, evaluation, discussion, and conclusions.

Chapter 1 – Introduction

This chapter introduces the overall CeylonRail Intelligence project and establishes the context of the individual AI-based railway fraud detection and passenger verification component. It presents the individual research problems, research gaps, research aim and objectives, research questions, scope, limitations, assumptions, individual contribution, and structure of the dissertation.

Chapter 2 – Literature Review

The second chapter presents a detailed review of existing research relevant to the individual research component. The chapter examines intelligent railway systems, railway ticketing technologies, railway ticket verification, fraud detection techniques, behavioral fraud detection, anomaly detection, machine-learning-based fraud detection, rule-based validation, passenger verification, QR-based validation, and related railway security approaches. The chapter also critically examines existing approaches to identify the research gap addressed by the proposed solution.

Chapter 3 – Research Methodology

The third chapter describes the research methodology adopted to develop and evaluate the proposed fraud detection and passenger verification component. It discusses the research approach, data requirements, data preparation, feature engineering, anomaly-detection model development, rule-based validation, risk scoring, passenger verification, evaluation criteria, and research workflow.

Chapter 4 – System and Solution Design

The fourth chapter presents the design of the proposed AI-based railway fraud detection and passenger verification solution. It describes the system architecture, data flow, module architecture, fraud detection process, feature structure, machine-learning

component, rule-based validation mechanism, risk scoring mechanism, passenger verification process, database interactions, and integration with the wider CeylonRail system.

Chapter 5 – Implementation and Testing

The fifth chapter presents the implementation of the proposed component. It describes the technologies and development environment used, implementation of the data-processing pipeline, anomaly-detection models, validation rules, risk assessment functionality, passenger and ticket verification functions, APIs, database interactions, user interfaces where applicable, and system integration. The chapter also describes functionality, integration, model, and performance testing.

Chapter 6 – Results and Validation

The sixth chapter presents the results obtained from the implementation and evaluation of the proposed component. It discusses the performance of the anomaly-detection models using relevant evaluation metrics such as precision, recall, F1-score, ROC-AUC, and false-positive rate. It also presents results related to risk scoring, system functionality, and integration testing.

Chapter 7 – Discussion

The seventh chapter interprets the research findings and discusses the significance of the results obtained. It compares the findings with relevant existing research, examines the effectiveness of the proposed approach, discusses the role of machine learning and rule-based validation, and considers the practical implications of the fraud detection and passenger verification component. The chapter also discusses limitations and factors that may affect the generalizability of the findings.

Chapter 8 – Conclusion and Future Directions

The final chapter summarizes the research undertaken and presents the major findings and contributions of the individual component. It discusses the extent to which the research objectives and research questions have been addressed, identifies the major limitations, and presents possible directions for future development. Future improvements may include larger and more diverse data sets, additional real-time railway data sources, improved anomaly-detection approaches, and further validation under actual railway operating conditions.

2. LITERATURE REVIEW AND RELATED WORK

2.1 Review Method and Scope

This chapter reviews existing research, technologies, and techniques related to railway ticket fraud detection and passenger verification. The review aims to understand current approaches, identify their limitations, and establish the research gap addressed by the proposed AI-Based Railway Fraud Detection and Passenger Verification component of CeylonRail.

The review focuses on five main areas: digital railway ticketing, passenger verification, behavioral fraud detection, machine-learning-based fraud detection, and transportation fare-evasion detection. Related research in financial fraud detection and smart transportation is also considered because these fields provide relevant approaches for identifying unusual behavior and fraudulent transactions.

A total of 16 studies were reviewed. Railway-focused studies include blockchain-based ticketing and QR validation [1], behavioral fraud detection [2], AI-based passenger verification [3], facial recognition [4], digital ticket verification and seat allocation [5][6], and smart ticketing systems [7][8]. Supporting studies examine passenger detection [9], railway passenger safety [10], public transportation fare evasion [11][14], fraud intention recognition [12], mobility irregularities [13], and general fraud detection using machine-learning and anomaly-detection techniques [15][16]. The literature was analyzed based on the techniques used, data characteristics, fraud or abnormality types, verification mechanisms, and applicability to the Sri Lankan railway environment. Attention was given to the integration of fraud detection, passenger verification, railway validation rules, risk assessment, and alert generation.

The Sri Lankan railway environment was also considered. Existing digital services, including smart ticketing and online reservation systems, demonstrate the use of digital ticket processing and passenger information. However, digital ticketing and conventional verification do not necessarily provide intelligent behavioral fraud detection. Therefore, the review focuses on whether existing approaches provide an integrated system capable of detecting suspicious behavior, verifying passenger-ticket relationships, applying railway-specific validation rules, generating risk assessments, and supporting operator alerts, particularly within the Sri Lankan railway context.

2.2 Domain and Theoretical Foundations

2.2.1 Digital Railway Ticketing

Digital ticketing provides opportunities for automated ticket management, validation, and passenger services. Farhan et al. [1] investigated blockchain-based railway ticketing using QR codes and smart contracts, demonstrating how digital technologies can improve ticket security and transaction management.

QR codes can provide unique ticket identifiers that can be checked against railway records. However, QR validation mainly determines whether a ticket is valid and does not necessarily detect misuse, duplication, or unusual passenger behavior. Therefore, digital validation represents an important foundation but is not sufficient as a complete fraud detection mechanism.

2.2.2 Passenger Verification

Passenger verification determines whether the person using a ticket is the legitimate passenger associated with it. Kirme et al. [3] investigated AI-based passenger verification, while Patil et al. [4] examined facial recognition for railway ticket verification. Gobhinath et al. [5] and Mhamane and Shriram [6] also explored digital ticket verification and passenger or identity-based railway systems.

These studies demonstrate that linking ticket information with passenger identity can strengthen verification. However, identity verification alone may not identify abnormal behavioral patterns. A legitimate passenger could still demonstrate suspicious purchase, cancellation, or ticket usage patterns. Therefore, the proposed system treats passenger verification and fraud detection as complementary functions. Verification focuses on the passenger-ticket relationship, while fraud detection analyzes behavioral and transactional patterns.

2.2.3 Smart Ticket Validation

Nawghare et al. [7] investigated smart railway ticket validation with alert mechanisms, while Reddy et al. [8] examined automated railway ticket management.

These studies demonstrate the value of automated validation and alert generation. However, predefined validation rules mainly identify conditions that have already been specified. They may not detect complex behavioral relationships across multiple transactions.

The proposed approach therefore combines rule-based validation for explicit railway requirements with machine-learning-based anomaly detection for behavioral fraud detection.

2.2.4 Behavioral Fraud Detection

Fraud may be identified through abnormal behavior rather than a single invalid transaction. Singh and Singh [2] examined fraud detection through customer behavior and activities, supporting the use of behavioral information for identifying suspicious activities.

For railway applications, relevant indicators may include:

- High ticket purchase frequency
- Repeated cancellations
- Duplicate ticket usage
- Route inconsistencies
- Unusual travel patterns

These indicators can be transformed into measurable features and analyzed together. Considering passenger history can provide additional information that may not be visible from an individual transaction.

2.2.5 Machine Learning for Fraud and Anomaly Detection

Machine learning enables systems to identify patterns within historical data and detect transactions that differ from expected behavior. In fraud detection, both supervised classification and anomaly-detection approaches can be applied depending on the availability and reliability of labelled fraud data.

Supervised classification methods require labelled examples of fraudulent and legitimate transactions. Such approaches can be useful when sufficient and reliable fraud labels are available. However, fraud datasets may contain relatively few fraudulent transactions compared with legitimate transactions, and new forms of fraud may not always be represented in historical labels. For this reason, the proposed research focuses on machine-learning-based anomaly detection rather than using a conventional supervised classifier as the primary detection method. Three approaches are considered: Isolation Forest, One-Class Support Vector Machine (One-Class SVM), and Autoencoder.

Isolation Forest identifies observations that are easier to isolate from most of the dataset and can therefore be used to identify potentially abnormal transactions. One-

Class SVM learns a boundary around normal observations and identifies observations that fall outside the learned region. Autoencoders learn to reconstruct the patterns present in the training data, with larger reconstruction errors potentially indicating anomalous transactions.

These approaches are appropriate for the research because fraudulent railway transactions may appear as deviations from normal transactions, passenger, fare, route, payment, booking, scanning, or inspection patterns. The three approaches also provide an opportunity to compare different anomaly-detection techniques under the same railway transaction dataset.

The fraud labels available in the dataset are used for evaluation of the detected anomalies, rather than being used as predictive input features. This allows the study to assess how effectively the anomaly-detection approaches identify transactions associated with known fraudulent cases.

2.2.6 Anomaly and Irregularity Detection

Anomaly detection focuses on identifying observations that differ significantly from normal patterns. This approach is particularly relevant to fraud detection because fraudulent activities may not always follow previously known patterns.

Wang et al. [13] investigated mobility irregularities using smart transit card data, demonstrating the potential of transportation data for identifying deviations from normal behavior.

Chettiar [16] examined Autoencoders, Local Outlier Factor, and Isolation Forest for fraud detection in imbalanced datasets. This is relevant because legitimate transactions generally outnumber fraudulent transactions in many fraud-detection datasets.

The proposed research applies this anomaly-detection concept to railway transaction data using Isolation Forest, One-Class SVM, and Autoencoder. The models analyze transaction-related features to identify potentially abnormal transactions. The known fraud labels are subsequently used to evaluate detection performance using measures such as precision, recall, F1-score, ROC-AUC, and false-positive rate.

This approach is particularly relevant to the railway environment because unusual combinations of transaction and passenger-related characteristics may indicate suspicious activity even when a transaction does not violate a single predefined rule.

2.3 Related Research

The reviewed research demonstrates that different components of railway fraud detection and passenger verification have already been investigated. However, these components are generally addressed separately.

Farhan et al. [1] focused on secure digital railway ticketing using QR codes and blockchain-based technologies. The approach demonstrates how digital ticket information can be secured and validated. However, its primary emphasis is ticketing security and transaction management rather than behavioral fraud detection.

Singh and Singh [2] focused on monitoring customer behavior and activities for fraud detection. This work is important to the proposed research because it establishes the relevance of behavioral patterns in identifying fraudulent activities. The concept can be adapted to railway passengers by analyzing their ticket purchase and travel behavior. Kirme et al. [3] investigated AI-based passenger verification. This provides a foundation for applying artificial intelligence to passenger authentication. However, verification and fraud detection remain different problems. A passenger may successfully pass identity verification while still exhibiting suspicious transaction behavior.

Patil et al. [4] investigated facial recognition for railway ticket verification. The study demonstrates the potential of biometric information for linking a passenger to a ticket. Nevertheless, biometric verification alone does not provide comprehensive behavioral fraud detection. Gobhinath et al. [5] proposed a prototype for digital railway ticket verification and seat allocation. The study demonstrates the potential for integrating digital ticket management with railway seat allocation.

Mhamane and Shriram [6] investigated railway ticket verification and dynamic seat allocation using identity information. This demonstrates another approach to strengthening the relationship between passenger identity and railway ticket information. Nawghare et al. [7] investigated an Indian Railways smart ticket validation system with an improved alert mechanism. Alerts are particularly relevant to the proposed system because a fraud detection model should provide useful outputs to railway staff rather than simply producing an internal detection result.

Reddy et al. [8] examined a fool-proof railway ticketing management system. This research contributes to the broader objective of improving reliability and reducing weaknesses in railway ticket management.

Gorane and Joshi [9] investigated passenger detection using machine-learning techniques for railway accident reduction. Although the main purpose of the research

is passenger safety rather than ticket fraud, it demonstrates the application of machine learning to railway passenger-related problems.

Alawad et al. [10] investigated the use of big data for enhancing passenger safety in railway stations. Their work highlights the value of large-scale passenger-related information for intelligent railway management. The same general principle can be applied to fraud detection, where multiple records can be analyzed to identify patterns.

Delfau et al. [11] investigated machine learning for tackling fare evasion in public transportation. This is directly relevant to railway fraud detection because fare evasion and ticket misuse both involve violations of transportation payment or ticketing requirements. López Herrera et al. [12] investigated deep-learning-based fraud intention recognition in a public transportation context. Their work demonstrates that transportation-related fraud can be investigated using advanced machine-learning techniques and behavioral information.

Wang et al. [13] investigated mobility irregularity detection using smart transit card data. This research is particularly relevant because transportation cards and railway ticket records can contain information about travel frequency, routes, and passenger movement patterns.

Sánchez-Martínez [14] examined fare noninteraction and evasion using disaggregate fare transaction data. The study demonstrates the usefulness of detailed transaction data for understanding fare-related behavior. Sadgali et al. [15] investigated human behavior scoring in credit card fraud detection. Although the financial domain differs from railway transportation, the principle of assigning a behavioral risk score is relevant to the proposed railway fraud detection system. Finally, Chettiar [16] investigated the combination of Autoencoders, Local Outlier Factor, and Isolation Forest for fraud detection in imbalanced datasets. This work provides useful evidence for the importance of anomaly detection and appropriate handling of imbalanced fraud data.

Overall, the reviewed studies demonstrate that the individual technologies required for intelligent railway fraud detection already exist in different research areas. However, they are distributed across separate systems and domains.

2.4 Existing Systems, Products and Techniques

Existing systems relevant to the proposed research can be classified into several categories: digital ticketing systems, QR-based ticket validation, identity-based verification, biometric verification, rule-based validation, alert systems, and machine-learning-based fraud detection.

2.4.1 Sri Lankan Railway Digital Services

Sri Lanka Railways already provides several digital railway services. The official Sri Lanka Railways website identifies services including the **Pravesha Smart Ticketing System** and the **Seat Reservations and E-Warrant System**.

The online reservation system includes user authentication, journey information, passenger information, and identity-related details. The system also requires NIC information for relevant passengers and provides mechanisms for verifying reservation information. The official railway reservation conditions also state that reserved tickets are associated with specified trains and that tickets are non-transferable. Railway officers have authority to request and check tickets.

These existing services demonstrate that Sri Lanka already has a foundation for digital railway ticket management and passenger-related verification. However, based on the reviewed literature and available official Sri Lanka Railways information, an integrated AI-based system specifically combining behavioral fraud detection, passenger verification, rule-based ticket validation, fraud risk scoring, and automated fraud alerts was not identified.

This distinction is important for the research gap. The proposed research is therefore not claiming that Sri Lanka has no digital ticketing or passenger verification. Instead, the gap concerns the absence of an identified integrated intelligent fraud detection layer that analyses passenger behavior and transaction patterns together with ticket and passenger verification.

2.4.2 QR and Digital Ticket Validation

QR-based ticketing provides a fast mechanism for identifying and validating tickets. Farhan et al. [1] demonstrated the use of QR codes within a secure railway ticketing framework.

The major advantage of QR-based validation is speed and convenience. However, the validation mechanism normally focuses on the ticket itself. Additional information is required to determine whether the passenger's usage behavior is suspicious. Therefore, QR or ticket information can serve as part of a wider railway fraud-detection framework, although the current proposed component focuses on the available transaction, passenger, route, fare, booking, scanning, and inspection information rather than implementing a separate biometric or QR-verification mechanism.

2.4.3 Identity and Biometric Verification

Identity-based systems can improve confidence that a ticket is being used by the appropriate passenger. Research by Kirme et al. [3], Patil et al. [4], and Mhamane and Shriram [6] demonstrates different approaches to passenger identity verification.

However, these systems primarily focus on establishing identity rather than analyzing long-term behavioral patterns. The proposed system therefore considers passenger verification as one component of a wider fraud detection process.

In the implemented component, passenger and ticket information are examined through railway-specific validation rules. The research does not claim implementation of facial recognition or biometric verification.

2.4.4 Rule-Based Validation

Rule-based validation is useful for conditions that can be explicitly defined. Examples include:

- checking whether a ticket exists;
- checking ticket validity;
- checking whether the ticket corresponds to the selected journey;
- checking whether the ticket has already been used;
- checking passenger-ticket information;
- checking whether a reserved ticket is being used for the correct train.

Such rules are important because railway requirements contain conditions that can be represented deterministically.

However, rule-based systems may have difficulty identifying previously unknown behavioral patterns. Therefore, the proposed system combines rule-based validation with machine-learning-based anomaly detection.

2.4.5 Machine-Learning-Based Fraud Detection

Machine learning provides an additional layer for identifying suspicious behavior. Research by Delfau et al. [11], López Herrera et al. [12], Wang et al. [13], Sadgali et al. [15], and Chettiar [16] demonstrates the usefulness of machine-learning and behavioral approaches for transportation or transaction fraud detection.

Unlike a conventional supervised classification approach, the proposed research uses **anomaly-detection techniques** to identify transactions that differ from learned patterns of normal activity.

Three machine-learning approaches are implemented:

1. **Isolation Forest** – identifies observations that can be isolated relatively quickly from the majority of observations.
2. **One-Class SVM** – learns the boundary of normal observations and identifies observations outside the learned boundary.
3. **Autoencoder** – learns to reconstruct normal transaction patterns and uses reconstruction error to identify potentially anomalous observations.

The models are applied to the railway transaction features available in the dataset. The fraud-related fields are retained for evaluation but are not used as predictive features. This allows the study to evaluate how effectively the anomaly-detection models identify known fraudulent transactions.

The resulting anomaly information is subsequently incorporated into the risk-scoring process so that potentially suspicious transactions can be categorized according to their level of risk.

2.4.6 Risk Scoring and Alerts

A practical fraud-detection system should provide outputs that can be understood and acted upon. Rather than treating every unusual activity as confirmed fraud, a risk score can be used to indicate the level of concern.

For example, a passenger with a normal travel pattern may receive a low-risk result, while a combination of duplicate ticket usage, unusual route behavior, and excessive cancellations may result in a higher risk score. The proposed component categorizes transactions into low-, medium-, and high-risk groups based on the calculated risk score. This provides a more interpretable output for railway staff and supports prioritization of potentially suspicious transactions for further inspection.

The alert-oriented concept is consistent with the ticket-validation approach investigated by Nawghare et al. [7]. However, the proposed component combines risk assessment with machine-learning-based anomaly detection and railway-specific validation rules.

2.5 Gap Synthesis and Conceptual Framework

2.5.1 Research Gap

The literature review identifies several important research gaps.

First, existing railway studies have largely focused on individual functions such as digital ticketing, ticket validation, passenger identification, biometric verification, or seat allocation [1], [3]–[8]. These technologies are useful but do not necessarily provide a complete behavioral fraud detection solution.

Second, transportation research has demonstrated the usefulness of machine learning for fare evasion, fraud intention recognition, and mobility irregularity detection [11]–[14]. However, these approaches are mainly studied in transportation environments outside Sri Lanka.

Third, general fraud-detection research has demonstrated the usefulness of behavioral scoring and anomaly detection [2], [15], [16]. However, these techniques have not been directly integrated with Sri Lankan railway ticketing rules and passenger verification requirements in the reviewed literature.

Most importantly, a Sri Lankan context-specific gap can be identified. Sri Lanka Railways already has digital services, including online seat reservations, e-ticket-related processes, “ **Pravesha**” smart ticketing, and passenger identification requirements. However, the reviewed research and available official information did not identify an existing Sri Lankan railway system that integrates:

1. railway ticket validation;
2. passenger verification;
3. historical passenger behavior analysis;
4. transaction-based fraud detection;
5. machine-learning-based fraud classification;
6. railway-specific rule-based validation;
7. fraud risk scoring; and
8. automated alerts for suspicious activities.

Therefore, the research gap is not simply the lack of digital railway ticketing in Sri Lanka. Instead, the gap is the lack of an identified integrated AI-based railway fraud detection and passenger verification framework specifically designed for the Sri Lankan railway environment. This creates an opportunity to adapt established techniques from railway ticketing, transportation analytics, behavioral fraud detection, and machine learning to the Sri Lankan railway context.

2.5.2 Summary of the Research Gap

Table 1- Summary of Research Gap

Area	Existing Research/System	Identified Limitation	Proposed Contribution
Digital ticketing	QR and digital ticketing [1]	Mainly validates ticket information	Combine ticket validation with fraud detection
Behavioral fraud detection	Customer behavior analysis [2]	Not railway-specific	Apply behavioral analysis to railway passengers
Passenger verification	AI/biometric verification [3], [4]	Mainly focuses on identity	Combine identity verification with behavioral fraud detection
Railway ticket verification	Digital verification and seat systems [5], [6]	Limited behavioral analysis	Integrate behavioral indicators
Smart ticket validation	Alert-based railway systems [7], [8]	Mainly validation-focused	Combine rules with ML-based risk assessment
Railway ML applications	Passenger detection and big-data applications [9], [10]	Different objectives	Apply ML specifically to ticket fraud
Public transport fraud	Fare evasion and fraud detection [11], [12]	Mainly outside Sri Lanka	Adapt concepts to Sri Lankan railway data
Mobility analysis	Transit-card irregularity detection [13]	Not specifically railway fraud	Use travel behaviour as fraud indicators
Fare transaction analysis	Fare evasion analysis [14]	Different transportation contexts	Incorporate transaction patterns
Behavioral scoring	Credit-card fraud [15]	Financial domain	Adapt risk scoring to railway passengers

Anomaly detection	Multiple ML anomaly techniques [16]	General fraud domain	Use findings to support railway fraud detection
Sri Lankan integration	Existing digital railway services	No identified integrated AI fraud-detection and passenger-verification framework in the reviewed sources	Develop a Sri Lankan railway-specific integrated framework

3. RESEARCH METHODOLOGY

3.1 Research Design

This research follows a quantitative and experimental research design to develop and evaluate an AI-based railway fraud detection and passenger verification component for the CeylonRail Intelligence system. The research is specifically designed for the Sri Lankan railway environment and focuses on identifying potentially fraudulent and abnormal railway ticket transactions through the analysis of passenger, ticket, journey, payment, scanning, and inspection-related information.

The research adopts a data-driven machine-learning approach. A railway ticket fraud dataset containing 9,800 records and 18 attributes is used as the main research dataset. The dataset contains information related to railway transactions from multiple years and includes a labelled fraud indicator and fraud type. The available data provides a basis for analyzing both normal and fraudulent railway transaction behavior.

The research investigates machine-learning-based anomaly detection because fraudulent activities may appear as unusual patterns within many legitimate railway transactions. Instead of relying exclusively on predefined fraud rules, the proposed approach analyses transaction characteristics to identify observations that differ from normal patterns.

Three machine-learning approaches are considered in the experimental study:

1. Isolation Forest
2. One-Class Support Vector Machine (One-Class SVM)
3. Autoencoder

Isolation Forest is investigated as the main tree-based anomaly-detection technique. One-Class SVM provides a boundary-based anomaly-detection approach, while the Autoencoder provides a neural-network-based approach based on reconstruction error.

The research uses the labelled fraud field as the ground-truth reference during evaluation. However, the fraud label is not treated as a direct predictive feature when the anomaly-detection models are trained. This allows the research to evaluate whether the models can identify suspicious transactions based on the available transaction characteristics.

The dataset contains transactions covering the period from 2020 to 2024. The experimental analysis shown in the implementation uses 2024 transactions as the test set, containing 2,750 records. This provides a temporally separated evaluation set that can be used to examine how well the developed models identify fraudulent transactions in a later period.

3.2 Data, Participants or Study Materials

3.2.1 Railway Fraud Dataset

The primary study material is a CSV dataset containing approximately 9,800 railway transaction records and 18 attributes relevant to railway ticketing and passenger activities.

Table 2 - Dataset Attributes

Attribute	Description
transaction_id	Unique transaction identifier
passenger_id	Passenger identifier
passenger_name	Passenger name
date	Transaction date
time	Transaction time
day_of_week	Day of the transaction
route	Railway journey route
travel_class	Travel class
ticket_type	Ticket type
standard_fare_lkr	Standard fare
fare_paid_lkr	Actual fare paid
payment_method	Payment method
booking_channel	Booking channel
seat_number	Assigned seat
ticket_scan_count	Number of ticket scans
inspection_status	Ticket inspection result
is_fraud	Fraud classification
fraud_type	Fraud category

The dataset contains numerical, categorical, temporal, transactional, and fraud-related information.

3.2.2 Dataset Structure

The dataset consists of 9,800 records and 18 columns. The fraud attribute provides the ground-truth fraud label, while fraud type identifies the relevant fraud category.

Passenger identifiers and names are treated as sensitive information and are excluded from predictive modelling and public research artefacts. Transaction identifiers are also excluded because they do not represent meaningful behavioral characteristics.

3.2.3 Data Period

The dataset covers transactions from 2020 to 2024. The records are ordered by date and time, with 2024 used as the test period, resulting in 2,750 test transactions.

Earlier transactions are used for model development. This time-based separation provides a more realistic evaluation by testing the models on a later period rather than randomly mixing transactions from different years.

3.2.4 Fraud Labelling

The is fraud attribute indicates whether a transaction is fraudulent, while fraud type provides additional classification information. These labels are used as ground truth during evaluation.

The fraud labels are not provided as predictive inputs to the anomaly-detection models. Instead, model predictions are compared with the actual labels to assess their ability to identify known fraudulent transactions.

3.3 Experimental or Study Procedure

The experimental process consists of data preparation, feature engineering, model development, evaluation, risk scoring, and passenger-ticket verification.

3.3.1 Data Loading and Initial Analysis

The CSV dataset is loaded into the analysis environment, and date and time fields are processed appropriately. Initial analysis examines dataset dimensions, data types, missing values, duplicates, fraud distribution, date range, and categorical consistency.

3.3.2 Data Pre-processing

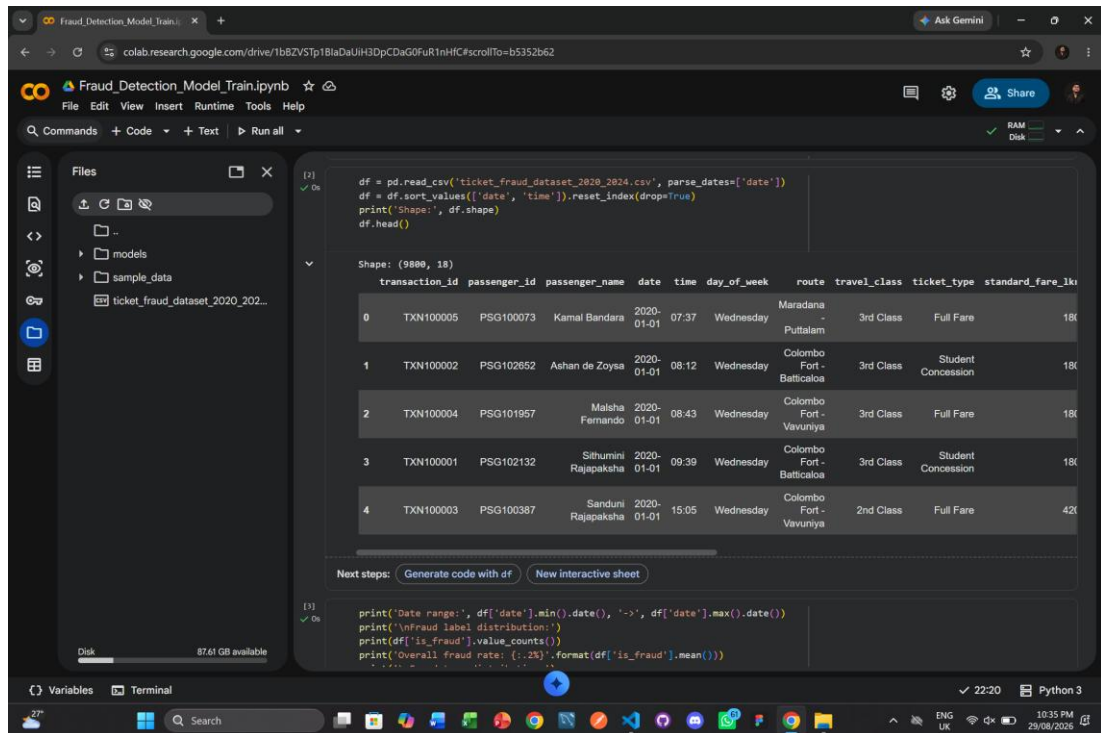


Figure 2 - Data Set Processing

Preprocessing includes handling missing or invalid values, checking duplicate records, and standardizing categorical and temporal information.

Identifying fields such as passenger name, passenger, and transaction_id are excluded from predictive modelling to protect privacy and avoid using non-behavioral identifiers.

3.3.3 Feature Selection and Engineering

Relevant transactional and behavioral characteristics are transformed into machine-learning features. These include:

- Booking frequency
- Ticket usage and scanning patterns
- Duplicate ticket usage
- Cancellation behavior
- Route inconsistencies
- Unusual travel behavior
- Inspection status
- Fare inconsistencies
- Passenger-ticket inconsistencies

The is fraud and fraud type fields are excluded from the model inputs because they represent the target outcome.

3.3.4 Training and Test Data

A time-based evaluation strategy is used, with 2024 transactions (2,750 records) reserved for testing and earlier transactions used for model development.

This approach simulates a practical scenario in which historical transactions are used to develop a model that is later applied to future railway transactions. Cross-validation is not used in the current implementation.

3.3.5 Isolation Forest

Isolation Forest is implemented as a tree-based anomaly-detection method. It identifies observations that can be isolated more easily from the remaining data. Transactions with unusual feature combinations may therefore receive higher anomaly scores.

The resulting predictions are compared with the is fraud labels in the 2024 test set.

3.3.6 One-Class SVM

One-Class SVM learns a boundary representing normal transaction behavior. Transactions falling outside this boundary are classified as potential anomalies.

Its predictions are evaluated against the same 2024 test dataset.

3.3.7 Autoencoder

An Autoencoder learns to reconstruct transaction features. Transactions that differ significantly from learned patterns may produce higher reconstruction errors.

A threshold is applied to the reconstruction error to identify potential anomalies, which are then evaluated using the labelled test data.

3.3.8 Model Comparison

The three models—Isolation Forest, One-Class SVM, and Autoencoder—are evaluated using the same 2024 test period.

Performance is assessed using measures including:

- Precision
- Recall
- F1-score
- ROC-AUC

- False-positive rate

The detailed numerical results are presented in the results chapter.

3.3.9 Risk Scoring

In addition to binary anomaly predictions, the system generates a risk score representing the estimated level of concern associated with each transaction.

For the 2,750 transactions in the 2024 test set, the current risk distribution is:

Table 3 - Risk level classification

Risk Level	Transactions	Percentage
Low	2,271	82.58%
Medium	328	11.93%
High	151	5.49%
Total	2,750	100%

The risk categories provide an additional mechanism for prioritizing transactions requiring further investigation.

3.3.10 Passenger and Ticket Verification

Fraud detection is complemented by rule-based passenger and ticket verification. Relevant information includes ticket details, passenger information, route, travel class, ticket type, scan count, inspection status, and fare information.

The rule-based component identifies explicit inconsistencies, while the machine-learning models identify potentially abnormal behavioral patterns. This combination provides both deterministic ticket validation and intelligent fraud detection.

3.4 Measures and Evaluation Metrics

The developed models are evaluated using quantitative measures. Since the test data contain known fraud labels, anomaly predictions can be compared with the is fraud ground truth.

3.4.1 Confusion Matrix

The confusion matrix consists of four outcomes:

- **True Positive (TP):** fraudulent transaction correctly identified as fraudulent/anomalous.

- **True Negative (TN):** legitimate transaction correctly identified as normal.
- **False Positive (FP):** legitimate transaction incorrectly identified as fraudulent/anomalous.
- **False Negative (FN):** fraudulent transaction incorrectly classified as normal.

The confusion matrix provides the basis for calculating precision, recall, F1-score, and false-positive rate.

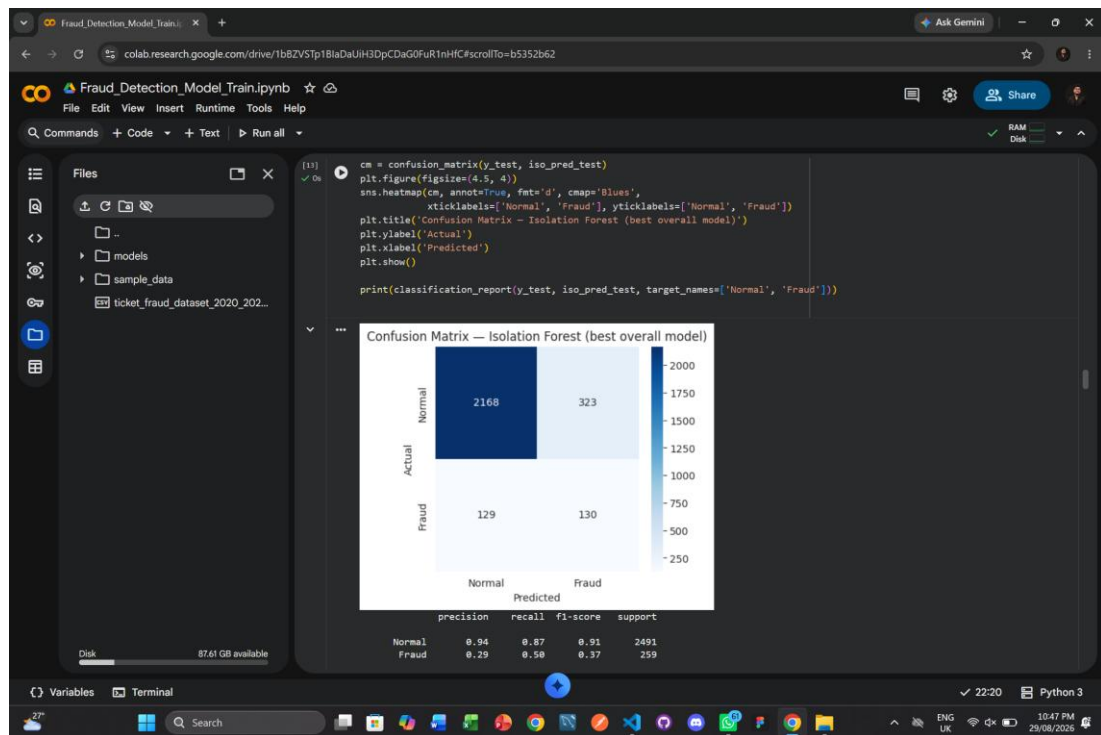


Figure 3 - Confusion Matrix

3.4.2 Precision

Precision measures the proportion of transactions identified as fraudulent that are fraudulent.

$$Precision = \frac{TP}{TP + FP}$$

It is important for reducing false fraud alerts.

3.4.3 Recall

Recall measures the proportion of actual fraudulent transactions correctly detected.

$$Recall = \frac{TP}{TP + FN}$$

High recall is important to minimize undetected fraud.

3.4.4 F1-Score

F1-score provides a balance between precision and recall.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

It is useful when both false positives and false negatives are important.

3.4.5 Accuracy

Accuracy measures the overall proportion of correctly classified transactions.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

It is considered alongside other metrics, particularly when the dataset is imbalanced.

3.4.6 False Positive Rate

False Positive Rate (FPR) measures legitimate transactions incorrectly identified as fraudulent.

$$FPR = \frac{FP}{FP + TN}$$

A low FPR helps reduce unnecessary alerts and staff workload.

3.4.7 ROC-AUC

Receiver Operating Characteristic Area Under the Curve (ROC-AUC) is used to evaluate the ability of model scores to distinguish fraudulent and legitimate transactions.

The ROC curve plots the true-positive rate against the false-positive rate across different decision thresholds.

The ROC-AUC value provides an overall measure of the separation between the two classes.

The experimental results show ROC curves for all three models, allowing their discriminatory performance to be compared.

3.5 Data Analysis

Data analysis is performed at both the dataset level and model-performance level.

3.5.1 Exploratory Data Analysis

The first stage analyses the characteristics of the railway dataset.

The analysis considers:

- transaction distribution;
- fraud distribution;
- fraud types;
- railway routes;
- travel classes;
- ticket types;
- payment methods;
- booking channels;
- ticket scan counts;
- inspection status;
- standard fares;
- paid fares; and
- temporal transaction patterns.

The purpose of this analysis is to understand the underlying characteristics of the dataset before applying anomaly-detection models.

3.5.2 Fraud Distribution Analysis

The fraud field is analyzed to determine the number and proportion of fraudulent and legitimate transactions.

The fraud_type field is also examined to understand the different categories represented in the dataset.

This analysis is useful for determining whether particular fraud types occur more frequently and whether some types may be more difficult for the anomaly-detection models to identify.

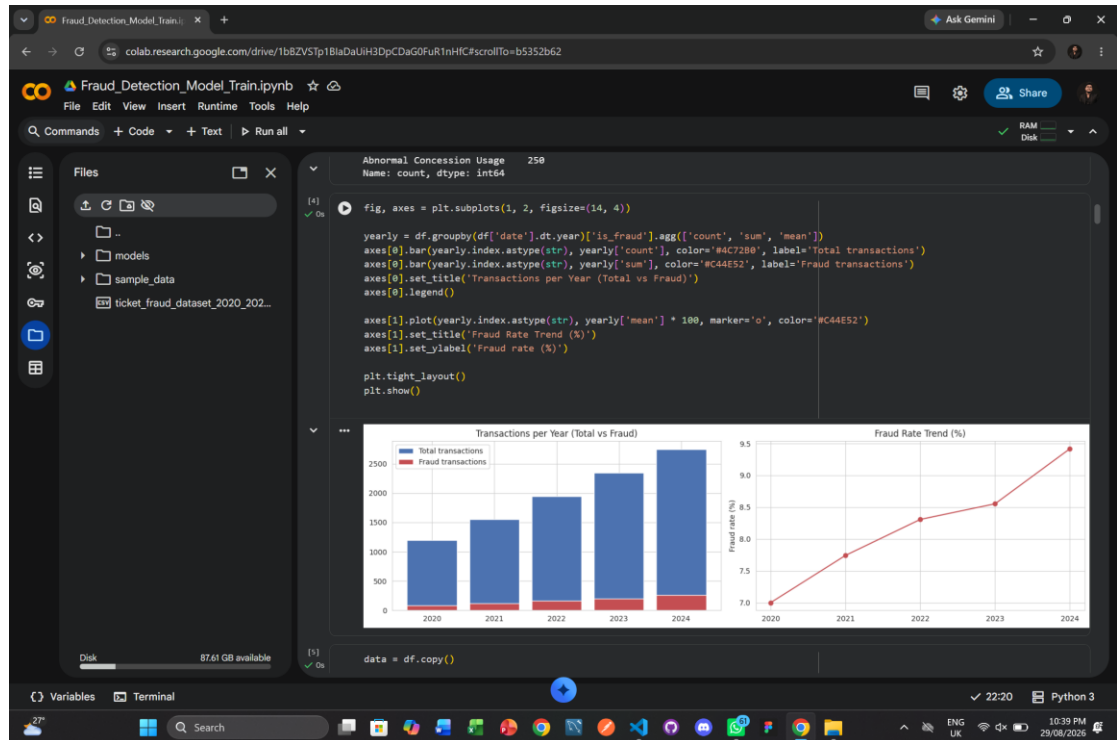


Figure 4 - Fraud Distribution Graph

3.5.3 Model Performance Analysis

The predictions from Isolation Forest, One-Class SVM, and Autoencoder are compared against the actual fraud labels.

The main comparison measures are:

- Precision;
- Recall;
- F1-score;
- ROC-AUC;
- False Positive Rate; and
- confusion matrix results.

The model-performance graph generated during the experiment demonstrates that the three models produce different precision, recall, F1-score, and ROC-AUC values.

The results indicate that Isolation Forest, One-Class SVM, and Autoencoder have different strengths. In particular, the displayed results show substantially higher recall

and ROC-AUC for One-Class SVM and Autoencoder than for Isolation Forest, while the Autoencoder has a slightly higher displayed F1-score than the other approaches.

The exact numerical values should be reported from the model output in the Results and Validation chapter rather than estimated from the graph.

3.5.4 Risk Distribution Analysis

The risk-score distribution is analyzed separately for normal and fraudulent transactions. The generated distribution shows that normal transactions are concentrated toward lower risk scores, while fraudulent transactions are more concentrated toward higher risk scores.

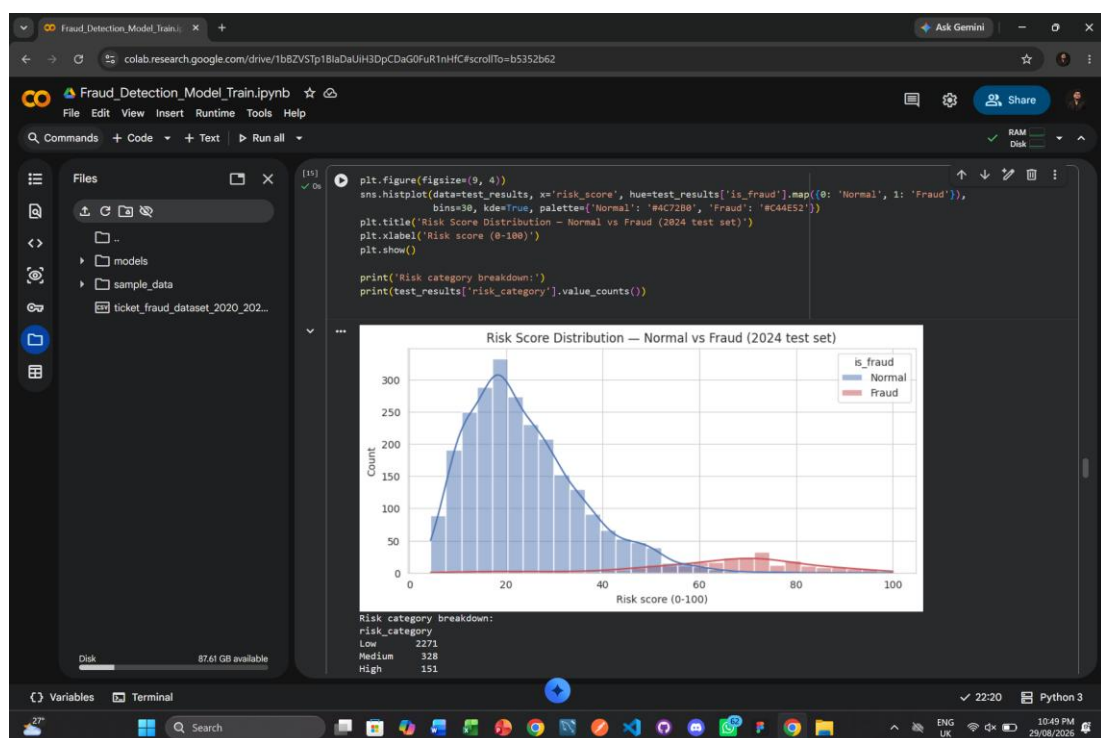


Figure 5 - Risk Score Distribution

This indicates that the risk score can provide useful information beyond a simple binary fraud prediction. The risk categories identified in the test set are:

- 2,271 low-risk transactions;
- 328 medium-risk transactions; and
- 151 high-risk transactions.

This analysis supports the use of risk-based outputs in the proposed railway fraud detection system. Analyzing these cases can help identify weaknesses in the selected features and improve future versions of the system.

3.6 Ethics, Privacy, Security and Responsible AI

The research involves railway passenger and transaction data; therefore, privacy, confidentiality, and responsible AI use are considered throughout the methodology.

3.6.1 Passenger Privacy and Data Confidentiality

Passenger identifiers such as passenger and passenger name should not be exposed in the dissertation, screenshots, tables, or research artefacts. Where possible, identifiers should be anonymized or pseudonymized. The original dataset should be securely stored, accessed only for research purposes, and not publicly shared without appropriate permission.

3.6.2 Responsible Fraud Detection

An anomaly does not necessarily indicate fraud, as unusual transactions may have legitimate explanations. Therefore, the proposed system is designed as a decision-support tool, not an autonomous system for declaring fraud. Suspicious transactions should be verified before operational action is taken.

3.6.3 False Positives, False Negatives and Explainability

Both false positives and false negatives are considered because incorrect detection may inconvenience legitimate passengers or allow fraudulent transactions to remain undetected. Where possible, the system should provide information about the abnormal patterns that contributed to an anomaly, supporting human investigation and interpretation.

3.6.4 Responsible Use of AI

The AI models are intended to support railway staff rather than replace human judgement. Model outputs should be considered alongside railway regulations, passenger verification, and other relevant evidence.

3.7 Reproducibility and Research Artefacts

Reproducibility is supported through documented preprocessing, feature engineering, model implementations, parameters, training and testing procedures, evaluation

scripts, and visualizations. The main models include Isolation Forest, One-Class SVM, and Autoencoder.

Due to the sensitive nature of passenger information, the original dataset should not be publicly distributed without permission. Instead, reproducibility can be supported through anonymized or synthetic datasets, source code, feature descriptions, model configurations, and documented procedures.

3.7.1 Experimental Configuration

The research should document the programming environment, libraries, software versions, model parameters, and experimental procedures. Historical railway transactions are used for model development, while the 2024 transaction records containing 2,750 observations are used as the test period.

3.7.2 Limitations of Reproducibility

Complete public reproducibility may be limited because the original railway dataset contains sensitive passenger and transaction information. Therefore, the research focuses on making the methodology, computational process, and evaluation procedures reproducible using an authorized or anonymized dataset.

4 . PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The AI-Based Railway Fraud Detection and Passenger Verification component is designed as one of the core intelligence components of the larger **CeylonRail Intelligence** platform. The overall project combines multiple intelligent railway functions, while this individual component focuses specifically on detecting suspicious railway ticket transactions, assessing fraud risk, and supporting passenger and ticket verification. The design requirements were derived from the identified railway fraud problem, the proposed CeylonRail system architecture, the available railway transaction dataset, and the requirements of the individual research component.

4.1.1 Functional Requirements

The system shall provide the following main functional capabilities.

FR1 – Railway Transaction Data Processing

The system shall accept railway transaction data containing passengers, ticket, journey, fare, payment, booking, scanning, and inspection-related information.

FR2 – Data Pre-processing

The system shall process raw railway transaction data by handling data types, missing values, categorical attributes, date/time information, and irrelevant identifying attributes before machine-learning analysis.

FR3 – Fraud Feature Generation

The system shall use relevant transactions and behavioral characteristics to identify potentially suspicious railway activities. These include abnormal transaction patterns, unusual ticket usage, route inconsistencies, ticket scanning behavior, fare-related inconsistencies, and other fraud indicators identified during the research.

FR4 – Anomaly Detection

The system shall analyze railway transactions using machine-learning-based anomaly-detection techniques.

The implemented research considers:

- Isolation Forest;
- One-Class SVM; and
- Autoencoder.

FR5 – Fraud Prediction

The system shall generate an anomaly/fraud-related prediction for each evaluated transaction.

FR6 – Fraud Risk Scoring

The system shall generate a risk score representing the level of concern associated with a transaction.

The risk output is categorized into:

- Low Risk;
- Medium Risk; and
- High Risk.

FR7 – Passenger and Ticket Verification

The system shall support validation of railway ticket and passenger-related information using the available transaction information and railway validation rules.

FR8 – Rule-Based Validation

The system shall support explicit railway validation conditions in addition to machine-learning-based anomaly detection.

Examples include checking for inconsistent ticket information, abnormal ticket usage, route inconsistencies, and inspection-related issues.

FR9 – Model Evaluation

The system evaluate the anomaly-detection approaches using appropriate performance metrics, including precision, recall, F1-score, and ROC-AUC.

FR10 – Fraud Alert Support

The system shall identify transactions that require further investigation based on their anomaly result and risk level.

The system is designed to support railway operators rather than automatically making a final decision regarding passenger fraud.

4.1.2 Non-Functional Requirements

In addition to functional requirements, the proposed component has several non-functional requirements.

- **Accuracy and Reliability**

The system should identify fraudulent and abnormal transactions with an acceptable level of detection performance while limiting false alerts.

- **Performance**

The system should be capable of processing railway transaction records efficiently so that fraud analysis can be performed without unnecessary delays.

- **Scalability**

The design should allow additional railway transactions to be processed as the system is expanded.

- **Security**

Passenger and transaction information must be protected against unauthorized access.

- **Privacy**

Personally identifiable information such as passenger names and identifiers should not be unnecessarily exposed during model development, testing, reporting, or system operation.

- **Maintainability**

The design should allow data-processing, machine-learning, risk-scoring, and verification components to be modified independently where possible.

- **Usability**

The outputs should be understandable to railway operators, particularly the risk category and supporting information associated with a suspicious transaction.

- **Interoperability**

The individual fraud-detection component should be capable of communicating with other components of the CeylonRail Intelligence platform through the overall system architecture.

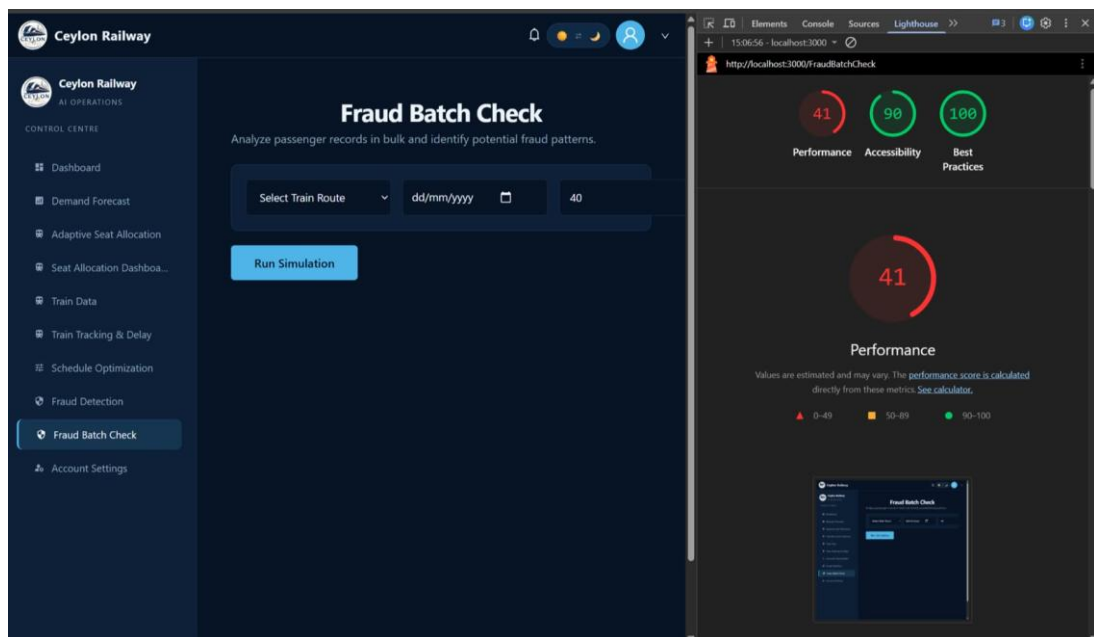


Figure 6 - Non-Functional Requirements

4.2 Component Architecture within the Complete Solution

The CeylonRail Intelligence project is designed as an integrated railway intelligence platform containing multiple specialised components. The overall architecture includes a User/Platform Access Layer, Core Intelligence Layer, Shared Railway Knowledge Layer, Data Services Layer, and Centralized Database Layer.

The individual fraud detection and passenger verification component operates mainly within the Core Intelligence Layer, while obtaining relevant information through the Data Services and Centralized Database layers.

4.2.1 Overall Ceylon Rail Architecture

The complete solution can be represented conceptually as:

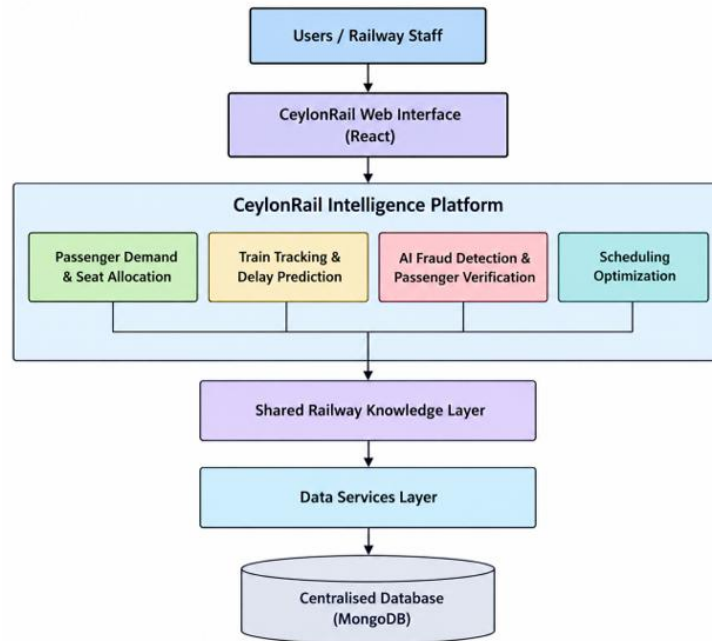


Figure 7 - System Diagram Ceylon rail

The individual research component communicates with the other CeylonRail components through shared data and platform services rather than functioning as an entirely independent railway system.

4.2.2 Fraud Detection Component Architecture

The detailed architecture of the individual component is:

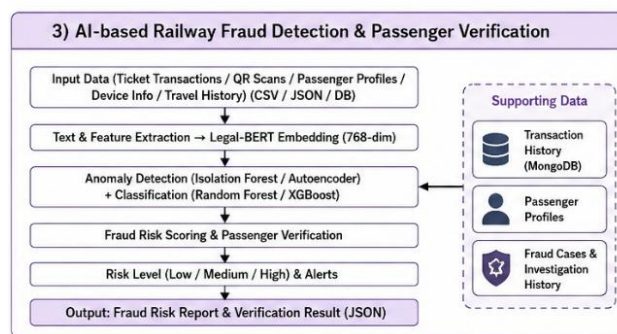


Figure 8 - Component Diagram

This architecture separates the major responsibilities of the component. Data processing prepares the input, machine-learning models perform anomaly detection, and rule-based verification provides railway-specific validation.

4.2.3 Interaction with Shared Railway Knowledge

The fraud-detection component requires railway-specific knowledge to interpret transaction information correctly.

The shared knowledge layer can provide information such as:

- railway ticketing rules;
- ticket validity requirements;
- route information;
- travel-class rules;
- fare-related rules;
- passenger verification requirements; and
- other operational constraints.

This allows the machine-learning output to be interpreted alongside explicit railway rules.

The combination is important because machine learning can identify unusual patterns, while railway rules can identify known conditions that are directly invalid or inconsistent.

4.2.4 Data Flow

The data flow through the component is as follows:

1. Railway transaction information enters the data-processing layer.
2. Data are cleaned and transformed.
3. Relevant features are generated.
4. The processed transaction is passed to the anomaly-detection models.
5. Each model generates an anomaly prediction or score.
6. The system evaluates the transaction against relevant railway validation rules.
7. Machine-learning and validation outputs contribute to the fraud-risk assessment.
8. A risk category is generated.
9. Suspicious transactions can be presented to railway operators for further investigation.

This design allows the component to function as an intelligent decision-support module within the larger CeylonRail platform.

4.3 Design Alternatives and Decisions

Several alternative approaches were considered during the design of the fraud-detection component.

4.3.1 Traditional Rule-Based Fraud Detection

One possible design is to implement the entire fraud detection process using predefined rules.

For example:

- IF ticket_scan count > expected limit
- THEN flag transaction

- IF route information is inconsistent
- THEN flag transaction

- IF ticket is invalid
- THEN reject transactions

The advantage of this approach is that the rules are easy to understand and explain. It can also be effective for clearly defined railway violations. However, a purely rule-based system has limitations when fraud patterns are complex or change over time. It may also fail to identify unusual combinations of multiple variables that were not explicitly defined in the rules. Therefore, rule-based validation is retained as a supporting mechanism rather than being used as the only fraud-detection approach.

4.3.2 Supervised Classification

A supervised classification approach could use labelled fraudulent and legitimate transactions to train a model such as Random Forest. This approach can be effective when a sufficiently representative set of labelled fraud examples is available.

However, railway fraud behavior can change over time, and previously unseen forms of suspicious behavior may not be represented adequately in the training labels. The research therefore focuses on anomaly detection, which allows the system to identify observations that differ from learned transaction patterns.

4.3.3 Isolation Forest

Isolation Forest was selected as one of the main approaches because it is specifically designed for anomaly detection.

Its main advantages include:

- suitability for anomaly detection;
- ability to handle multiple transaction features;
- relatively efficient processing;
- ability to produce anomaly scores; and
- suitability for identifying unusual observations.

It is therefore particularly relevant to transaction-level railway fraud analysis.

4.3.4 One-Class SVM

One-Class SVM was selected as an alternative because it provides a different anomaly-detection mechanism. Instead of using random tree isolation, the algorithm learns a boundary representing normal observations.

This provides a useful comparison against Isolation Forest and allows the research to determine whether a boundary-based approach provides better separation between normal and suspicious railway transactions.

4.3.5 Autoencoder

An Autoencoder was selected as a neural-network-based alternative.

The Autoencoder learns to reconstruct the input data. Transactions that are substantially different from the learned patterns can generate higher reconstruction errors. This provides a different approach to anomaly detection and is useful for comparing conventional machine-learning and neural-network-based approaches.

4.3.6 Final Design Decision

The final research design therefore uses three anomaly-detection approaches:

Table 4 - Model Comparison

Approach	Role
Isolation Forest	Primary tree-based anomaly-detection approach
One-Class SVM	Alternative boundary-based anomaly detector

Autoencoder	Neural-network-based anomaly detector
Rule-based validation	Railway-specific verification support

The models are compared experimentally rather than assuming that one technique will always provide the best performance.

The actual experimental results presented in the research show that the three approaches have different performance characteristics. Therefore, the final model selection is based on the evaluation results rather than solely on theoretical expectations.

4.4 Standards and Professional Practice

The development of the fraud-detection and passenger-verification component follows general software-engineering, data-science, security, privacy, and responsible-AI practices.

4.4.1 Data Management Practice

The research follows a structured data-processing process in which raw data are separated from processed analytical data.

The original dataset should be preserved securely, while derived datasets should be documented according to the preprocessing operations applied.

Data fields should have clear definitions, and transformations should be recorded so that the analysis can be reproduced.

4.4.2 Software Engineering Practice

The system should be designed using modular components.

The major modules include:

- data processing;
- feature engineering;
- anomaly detection;
- risk scoring;
- rule-based validation;
- passenger/ticket verification; and
- evaluation.

This modular approach improves maintainability and allows individual components to be modified without redesigning the entire system.

4.4.3 Machine-Learning Practice

The research follows appropriate machine-learning practices by separating model inputs from evaluation labels.

The is fraud label is used as ground truth for evaluation rather than being directly supplied as a predictive feature to the anomaly-detection models. The temporal test design using 2024 transaction records also helps reduce the risk of evaluating the model on the same period used for model development. Model performance is evaluated by using multiple metrics rather than relying on a single measure.

4.4.4 Privacy and Data Protection

Passenger information requires careful handling because the dataset contains potentially identifiable information. Passenger names and identifiers should not be unnecessarily exposed. For demonstrations and dissertation figures, anonymized identifiers should be used. Where research artefacts are shared, the original dataset should only be distributed if the necessary permission and data-sharing conditions are satisfied.

4.4.5 Security Practice

Access to the railway dataset and model outputs should be restricted to authorized users. Sensitive files should be stored securely, and unnecessary copies should be avoided. The system should also avoid exposing sensitive passenger information through error messages, logs, screenshots, or exported reports.

4.4.6 Responsible AI

The fraud detection component follows a human-in-the-loop approach. The machine-learning system identifies suspicious patterns, but the final operational decision should remain with an authorized railway officer or responsible human decisionmaker. This is important because an anomaly does not necessarily mean that a passenger has committed fraud. The system should therefore provide risk and supporting information rather than presenting every model prediction as a definitive conclusion.

4.4.7 Transparency and Explainability

The system should maintain sufficient information about how a transaction was identified as suspicious.

Possible supporting indicators include:

- unusual scan count
- unusual transaction pattern
- route inconsistency
- fare inconsistency
- inspection-related issue or
- high anomaly score.

Providing supporting information makes the system more useful to railway operators and allows suspicious cases to be reviewed before action is taken.

4.4.8 Testing and Validation

The component should be tested at multiple levels:

Data-level testing:

Verifies that input data are loaded and processed correctly.

Model-level testing:

Evaluates the anomaly-detection models using the test dataset.

Functional testing:

Checks risk scoring, validation, and output generation.

Integration testing:

Ensures that the fraud component can exchange relevant information with other CeylonRail components.

Security and privacy testing:

Checks that sensitive passenger information is not unnecessarily exposed.

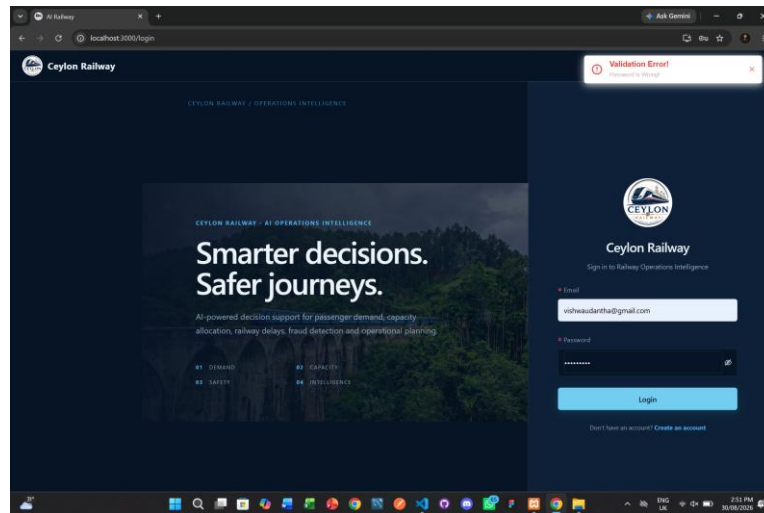


Figure 9 - validation error

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development Environment and Technology Stack

The AI-Based Railway Fraud Detection and Passenger Verification component was implemented as part of the CeylonRail Intelligence platform. The development process used Python for machine-learning and data-processing activities, while React and Flask were used to provide the application interface and communication between the user interface and backend services. MySQL was used for storing applications and transaction-related information.

The development environment was selected to support data processing, machine-learning experimentation, system integration, and application development.

5.1.1 Programming Language

Python was used as the primary programming language for machine-learning and data-processing components.

Python was selected because it provides extensive support for:

- data preprocessing;
- numerical computation;
- machine learning;
- neural networks;
- data visualization; and
- experimental model development.

The fraud-detection models, feature-processing procedures, evaluation procedures, and risk-scoring functionality were implemented using Python.

5.1.2 Development Environment

Google Colab was used as the primary environment for developing and evaluating machine-learning components.

The experimental work was performed using the CPU environment available through Google Colab. The environment provided an interactive platform for:

- loading the railway transaction dataset;
- preprocessing the data;
- performing exploration analysis;
- developing machine-learning models;

- evaluating model performance;
- generating ROC curves;
- generating risk-score distributions; and
- comparing the three anomaly-detection approaches.

The use of an interactive notebook environment also allowed the research experiments to be executed in a step-by-step manner.

5.1.3 Data Processing Libraries

Pandas were used for loading, manipulating, cleaning, and analyzing the railway transaction dataset. The dataset was loaded from a CSV file and processed into a structured data frame. Pandas were also used to examine dataset dimensions, transaction dates, categorical information, and fraud labels. NumPy was used for numerical processing and preparation of data for machine-learning operations.

5.1.4 Machine-Learning Libraries

Scikit-learn was used for implementing machine-learning and evaluation functionality.

The library was used for the anomaly-detection approaches including:

- Isolation Forest;
- One-Class SVM; and
- performance evaluation.

Scikit-learn also supported preprocessing and calculation of evaluation measures used during the experiments.

5.1.5 Deep-Learning Framework

TensorFlow was used to implement the Autoencoder-based anomaly-detection approach. The Autoencoder learns a representation of the transaction features and attempts to reconstruct the original input. Reconstruction error is then used as an indicator of potentially abnormal transactions.

5.1.6 Data Visualization

Matplotlib was used to generate graphical representations of the experimental results.

The visualizations include:

- model-performance comparisons;

- ROC curves;
- risk-score distributions; and
- other experimental analysis graphs.

These visualizations were used to support the comparison and interpretation of the machine-learning models.

5.1.7 Frontend

The application frontend was developed using React.

React provides the user-facing interface through which authorized users can interact with the fraud-detection functionality.

The frontend is responsible for presenting information such as:

- transaction information;
- verification results;
- fraud-related outputs;
- risk scores;
- risk categories; and
- relevant system responses.

The use of component-based frontend architecture allows different parts of the CeylonRail platform to be presented through a consistent interface.

5.1.8 Backend

Flask was used as the backend framework. The Flask backend acts as the communication layer between the React frontend, database, and fraud-detection functionality. The backend is responsible for processing requests received from the frontend, accessing relevant data, executing the required processing or model functionality, and returning results to the frontend.

The conceptual communication flow is:

5.1.9 Database

MySQL was used as the database technology for the application. MySQL provides storage for the application information required by the integrated system. The database layer supports the separation of data storage from the machine-learning processing layer and allows the backend to retrieve and update relevant information.

5.1.10 Version Control

GitHub was used for source-code version management.

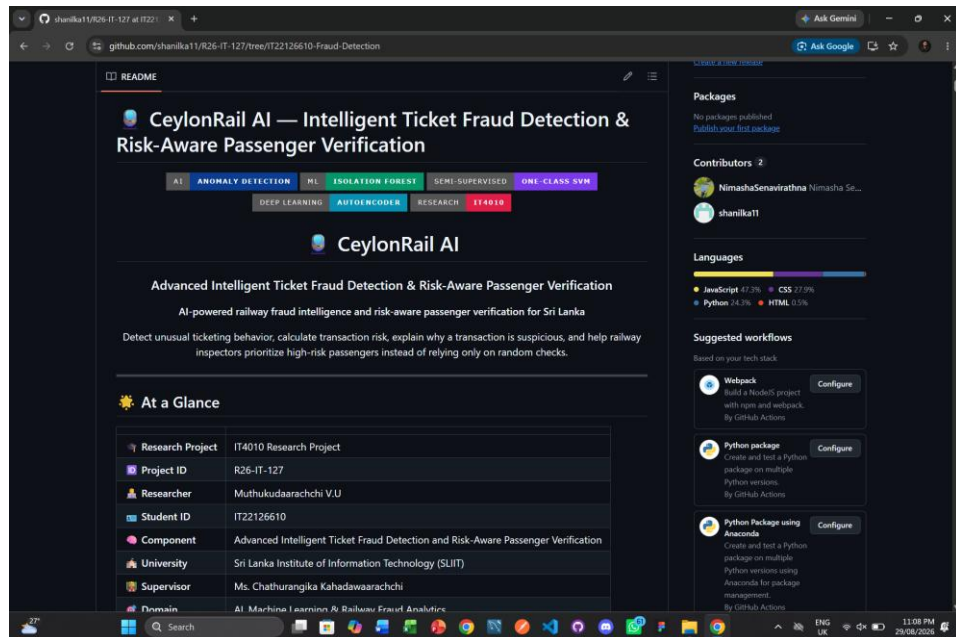


Figure 10 - Git Hub Branch of the component

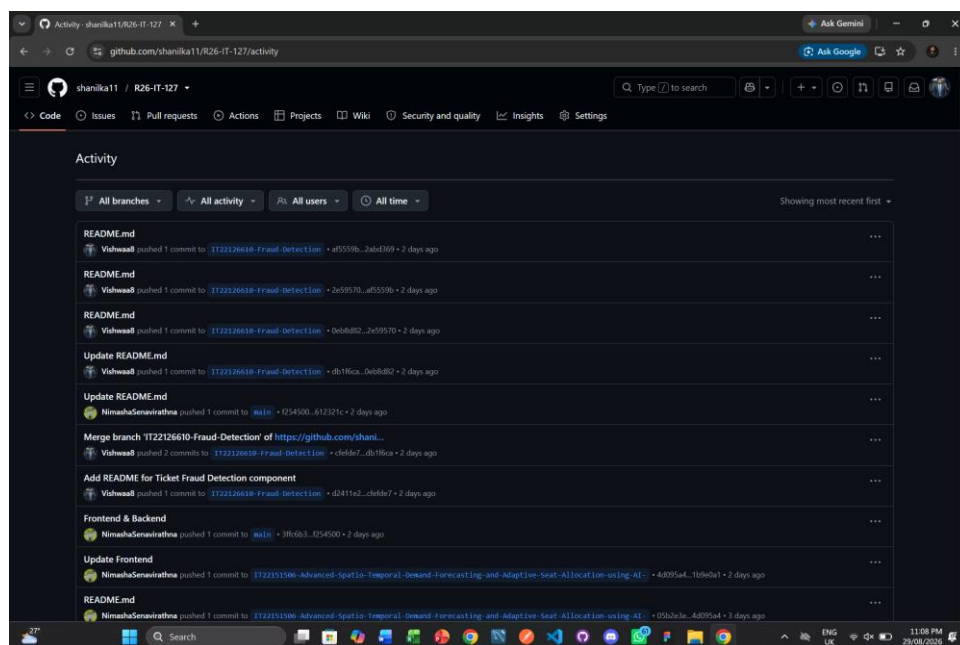


Figure 11 - Commit history

Version control supports:

- maintaining development versions;
- tracking code changes;
- maintaining the project source code;
- collaboration between group members; and
- maintaining an organized software-development process.

Because the project involves multiple individual components, version control is particularly useful for coordinating development between the different components of CeylonRail Intelligence.

5.2 Component Implementation

The fraud-detection component consists of data processing, feature engineering, anomaly detection, risk scoring, passenger/ticket verification, and application output.

5.2.1 Dataset Loading and Preprocessing

The railway transaction dataset was loaded from CSV using Pandas. It contains 9,800 records and 18 attributes, including transaction, passenger, route, fare, ticket, payment, scanning, inspection, and fraud information.

The data was processed by handling date/time information, categorical and numerical attributes, missing or inconsistent values, and irrelevant identifiers. Passenger identifiers were excluded from predictive features. The `is_fraud` and `fraud_type` fields were retained for evaluation and analysis but were not used as model inputs.

5.2.2 Feature Preparation

Transaction information was converted into model-ready features representing fare, route, travel class, ticket type, payment method, booking channel, scanning behaviour, inspection status, and temporal characteristics. The same processing procedure was applied to development and test data.

5.2.3 Anomaly Detection Models

Three anomaly-detection models were implemented:

- **Isolation Forest** – identifies observations that are easier to isolate from the normal transaction population.

- **One-Class SVM** – learns the normal transaction distribution and identifies observations outside the learned boundary.
- **Autoencoder** – reconstructs transaction features and uses reconstruction error to identify potential anomalies.

The models generated anomaly predictions and scores for subsequent risk analysis.

5.2.4 Passenger and Ticket Verification

The component also considers ticket, passenger, route, class, fare, scanning, and inspection information for verification. Machine-learning outputs can therefore be considered together with railway-specific validation information rather than relying solely on the model prediction.

5.2.5 Flask API and React Interface

The Flask backend provides the API between the machine-learning functionality, MySQL, and React frontend. The React interface displays transaction information, verification status, anomaly/fraud status, risk score, and risk category to the user.

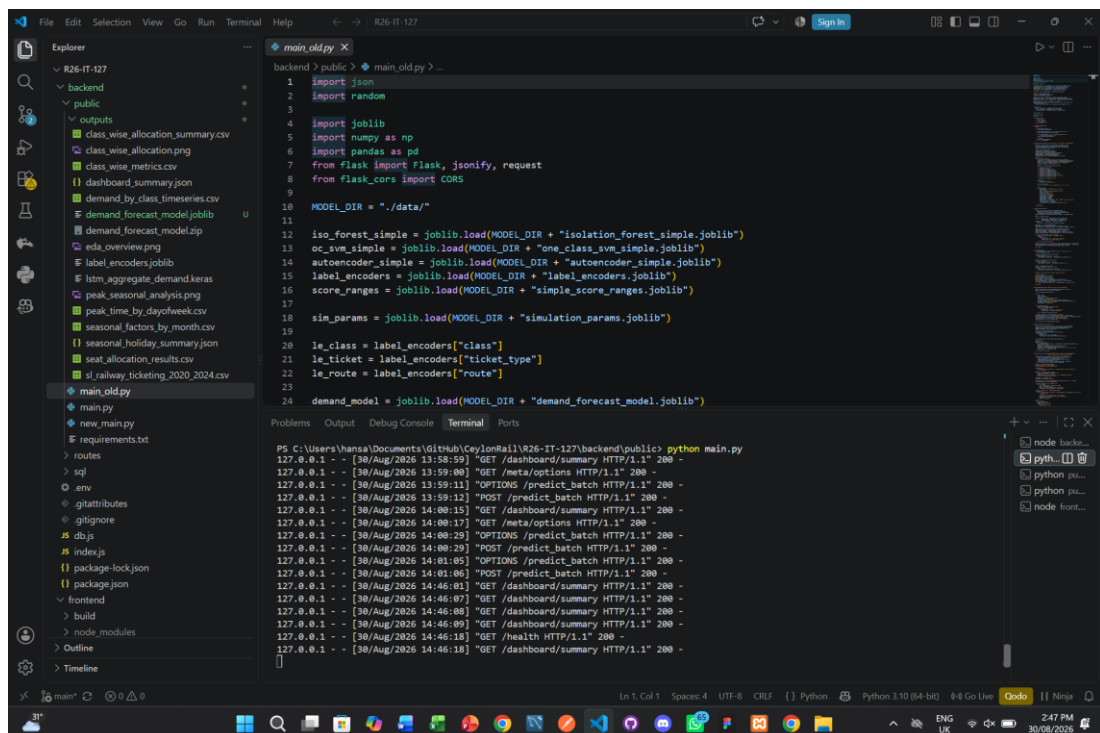


Figure 12 - working Backend

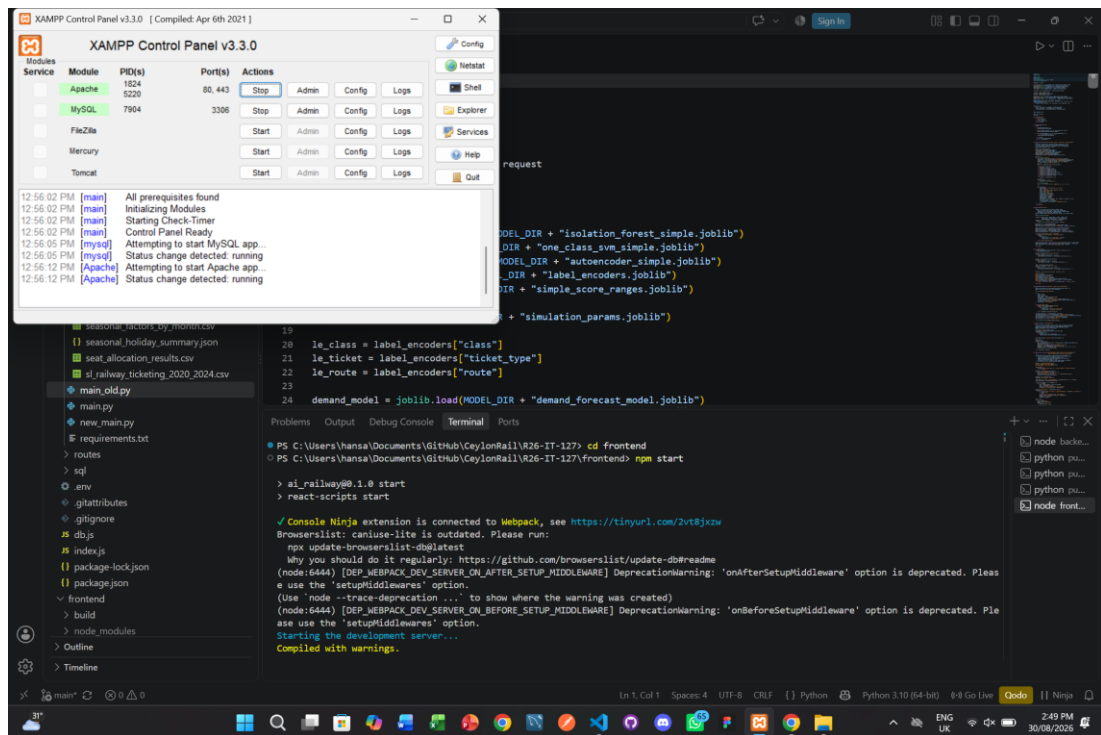


Figure 13 - Frontend starting

5.3 Integration with Other Components

The fraud detection component is integrated into the wider **CeylonRail Intelligence System**, which includes demand forecasting and seat allocation, train tracking and delay prediction, fraud detection and passenger verification, and train scheduling.

The shared architecture allows the components to exchange relevant railway information while maintaining their individual functions. Passenger, ticket, route, timing, and operational information can provide additional context for fraud verification and other railway intelligence tasks.

5.4 Verification and Software Testing

Testing was conducted at both machine-learning and application levels.

5.4.1 Model Testing

Isolation Forest, One-Class SVM, and Autoencoder were tested using the 2,750-record 2024 test set. Precision, Recall, F1-score, and ROC-AUC were used to compare model performance.

5.4.2 ROC and Risk-Score Testing

ROC curves were generated to compare the models' ability to distinguish fraudulent and legitimate transactions. Risk-score testing also confirmed the generation of Low, Medium, and High-risk categories and showed greater concentration of fraudulent transactions at higher risk levels.

5.4.3 Functional Testing

The main application functions tested included:

- dataset loading and preprocessing
- feature generation
- model execution
- risk scoring
- ticket verification
- React–Flask communication
- MySQL access; and
- frontend result display

5.4.4 API, Database and Integration Testing

API testing verified communication between React and Flask. Database testing verified MySQL connectivity and data retrieval. Integration testing confirmed that the fraud component operates correctly within the wider CeylonRail platform.

5.4.5 Testing Limitations

The testing focused on model performance and application functionality. Large-scale production load testing, extensive user acceptance testing, and long-term operational monitoring were outside the research scope.

5.5 Deployment and Operation

The completed system currently operates as a local research prototype. Machine-learning experiments were conducted using Google Colab CPU, while the integrated React, Flask, and MySQL application is operated locally.

5.5.1 Deployment Architecture

React Frontend → Flask Backend → MySQL + Fraud Detection Module

The front end communicates with the Flask API, which processes requests, retrieves required data, performs fraud detection and verification, and returns results to the front end.

5.5.2 Operational Workflow

The operational process is:

1. User accesses the React application.
2. A railway transaction is selected or submitted.
3. React sends the request to Flask.
4. Flask retrieves the required data from MySQL .
5. Fraud detection and verification are performed.
6. Risk and anomaly results are generated.
7. Results are returned to React and displayed to the authorized user.
8. Suspicious transactions can be reviewed by an operator.

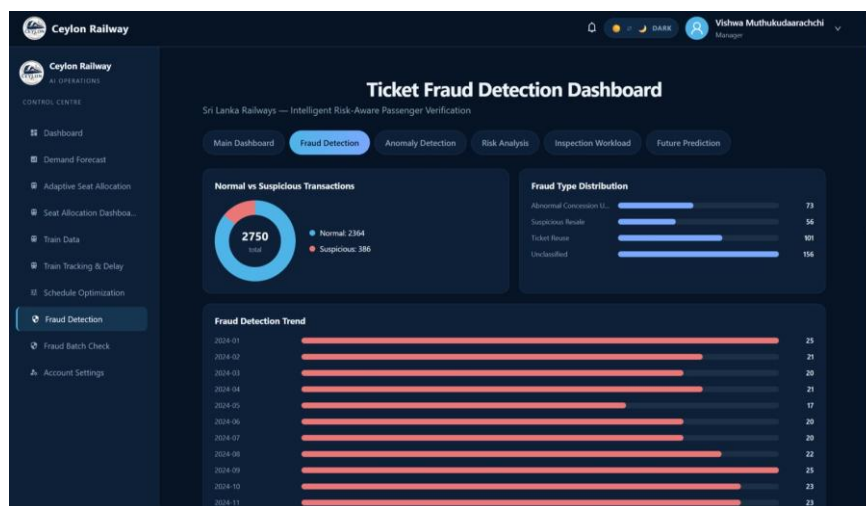


Figure 14 - React Frontend Dashboard

5.5.3 Source-Code Management

The project is maintained using GitHub for version control and collaboration. Confidential passenger data, credentials, passwords, and API keys should not be stored in the repository.

5.5.4 Future Deployment

The current system can be extended to a production environment using hosted services, secured APIs, authentication, managed databases, model-serving infrastructure, monitoring, and automated model updates. However, production deployment is outside the current research scope.

5.5.5 Operational Considerations

A production system would require secure access control, passenger-data protection, continuous monitoring, audit logging, backups, model-performance monitoring, and human review of suspicious transactions. The current implementation provides a technical research foundation rather than a production-ready railway system.

6. RESULTS AND VALIDATION

6.1 Research Results

The AI-Based Railway Fraud Detection and Passenger Verification component was evaluated using the developed railway transaction dataset and the implemented anomaly-detection models. The objective of the evaluation was to determine whether machine-learning techniques could identify abnormal and potentially fraudulent railway transactions and generate meaningful risk information.

The dataset used for the research contained 9,800 railway transaction records and 18 attributes. The available transaction information included passenger details, journey information, ticket information, fare information, payment method, booking channel, ticket scanning behaviour, inspection status, fraud labels, and fraud type.

The dataset contained labelled fraud information, allowing the model outputs to be compared against known transaction classifications during evaluation.

6.1.1 Experimental Dataset

The research used historical Sri Lankan railway transaction data covering the period from 2020 to 2024.

The data were organised chronologically before the modelling process. The 2024 records were used as the test set, providing 2,750 transactions for the final model evaluation.

The use of a separate test period provides a more realistic evaluation because the model is evaluated against transactions from a later period rather than simply evaluating the same observations used during model development.

6.1.2 Model Comparison

Three anomaly-detection approaches were implemented and compared:

1. Isolation Forest;
2. One-Class SVM; and
3. Autoencoder.

The models were selected because they provide different approaches to detecting abnormal transaction behavior.

Isolation Forest identifies observations that can be isolated relatively easily from the rest of the dataset. One-Class SVM learns a boundary representing the expected transaction population. The Autoencoder learns to reconstruct transaction features and uses reconstruction error as an indication of abnormality.

The experimental comparison produced the following results:

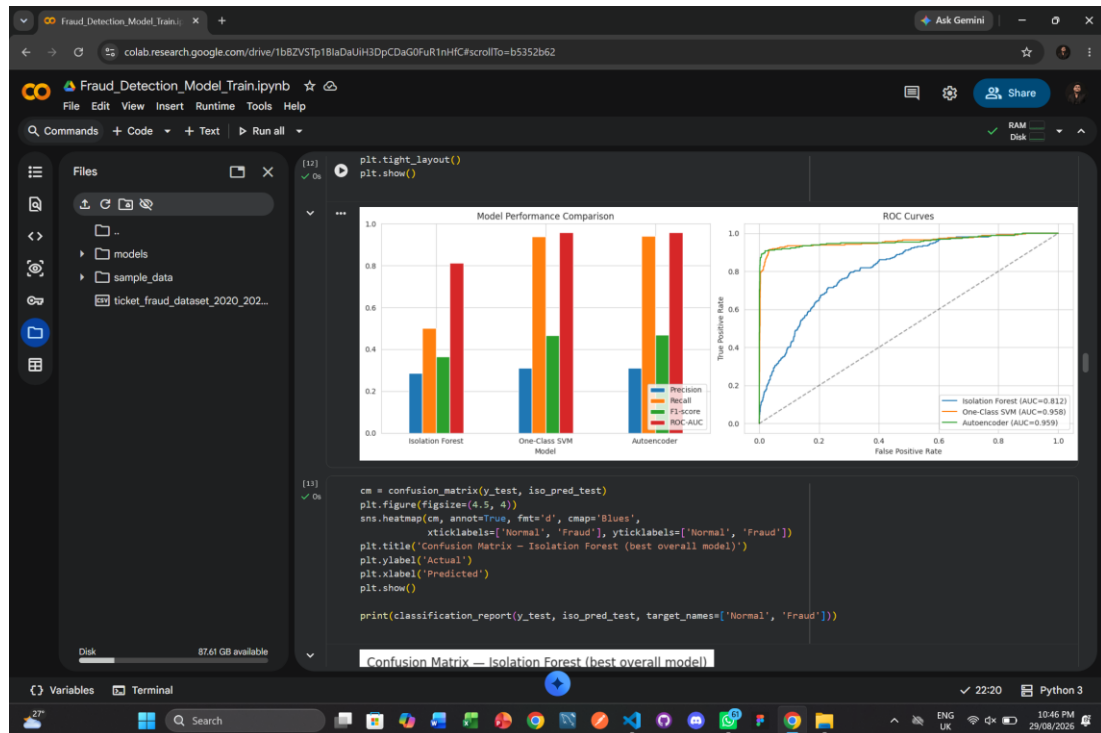


Figure 15 - Model Performance Results

Table 5 - Model Performance Comparison

Model	Precision	Recall	F1-Score	ROC-AUC
Isolation Forest	0.28	0.50	0.36	0.81
One-Class SVM	0.31	0.94	0.46	0.96
Autoencoder	0.31	0.94	0.47	0.96

The results demonstrate that the three approaches behave differently when detecting potentially fraudulent transactions. Isolation Forest achieved a ROC-AUC of approximately **0.81**, indicating that the model was able to distinguish between normal and fraudulent transactions to a useful degree. However, its recall was lower than that of the other two approaches.

One-Class SVM achieved a recall of approximately **0.94** and a ROC-AUC of approximately **0.96**. This indicates that it detected a large proportion of the labelled fraudulent transactions in the test set. The Autoencoder produced a similar recall and

ROC-AUC, with an F1-score of approximately **0.47**. This indicates that the neural-network-based approach was also capable of identifying a substantial proportion of the suspicious transactions. The results therefore demonstrate that anomaly-detection techniques can be applied to the available Sri Lankan railway transaction data to identify potentially abnormal passenger and ticketing behavior.

6.1.3 Precision, Recall and F1-Score Analysis

Precision measures the proportion of transactions identified as fraudulent that were actually fraudulent, while recall measures the proportion of known fraudulent transactions that were detected.

- The Isolation Forest achieved a precision of approximately 0.28 and recall of approximately 0.50.
- The One-Class SVM achieved a precision of approximately 0.31 and recall of approximately 0.94.
- The Autoencoder achieved a precision of approximately 0.31 and recall of approximately 0.94.

The higher recall of One-Class SVM and Autoencoder is particularly relevant for a fraud-detection research context because failing to identify fraudulent activity can result in undetected revenue loss or misuse of railway tickets. However, the precision values also indicate that the models generate false positives. Therefore, the anomaly-detection result should not be treated as automatic proof that a passenger has committed fraud.

6.1.4 ROC-AUC Analysis

ROC curves were generated to further compare the three anomaly-detection models.

The ROC curve evaluates the relationship between the true-positive rate and false-positive rate across different classification thresholds. The experimental results show that the One-Class SVM and Autoencoder achieved ROC-AUC values of approximately **0.96**, while Isolation Forest achieved approximately **0.81**. The higher ROC-AUC values indicate that One-Class SVM and Autoencoder provided stronger separation between normal and fraudulent observations within the evaluated test dataset. The ROC analysis therefore provides additional evidence supporting the performance differences observed in the precision, recall, and F1-score results.

6.2 Product and Technical Results

The research resulted in a functional fraud detection and passenger-verification component that can operate as part of the wider CeylonRail Intelligence platform.

6.2.1 Functional Product Results

The implemented component provides the following major capabilities:

- railway transaction data processing;
- fraud-related feature preparation;
- anomaly detection;
- comparison of multiple machine-learning models;
- fraud-risk scoring;
- risk categorization;
- passenger/ticket verification support;
- result visualization; and
- integration with the application backend.

The component therefore provides both an experimental machine-learning solution and an application-level mechanism for presenting the resulting information.

6.2.2 Machine-Learning Results

The technical implementation successfully executed three different anomaly-detection approaches.

The three models produced:

- anomaly predictions;
- evaluation scores;
- ROC curves; and
- risk-related outputs.

This demonstrates that the proposed solution is not limited to a single machine-learning technique and provides a basis for comparative evaluation.

6.2.3 Application Results

The application architecture uses:

- React for the frontend;
- Flask for backend services;

- MySQL for data storage;
- Python for machine-learning processing;
- scikit-learn for conventional machine-learning models; and
- TensorFlow for the Autoencoder.

The frontend provides an interface through which the results can be presented to an authorized user, while the Flask backend provides communication between the interface, data storage, and analytical functionality.

6.2.4 Risk-Based Output

One of the main technical outcomes of the research is the transformation of model outputs into an understandable risk representation.

Instead of presenting only:

- Normal / Fraud

the system provides:

- Risk Score → Risk Category → Further Investigation

This provides a more flexible approach for railway operators because suspicious transactions can be prioritized according to their risk level.

For example, a high-risk transaction can be examined before a low-risk transaction rather than treating all anomalies identically.

6.2.5 Technical Contribution

The technical contribution of the component is the combination of:

Railway transaction data + anomaly detection + risk scoring + railway-specific verification

This is particularly relevant to the Sri Lankan railway context because the component is designed around railway transaction information and operational requirements rather than being a generic financial fraud-detection system.

6.3 User or Stakeholder Validation

User and stakeholder validation is an important part of evaluating an operational railway system. However, the current research primarily focused on technical and experimental validation.

A formal large-scale stakeholder survey, passenger study, or controlled railway-operator user study was not conducted as part of the implemented research.

Therefore, numerical user-satisfaction claims are not presented.

6.3.1 Technical Stakeholder Perspective

The system was designed with railway operators as the primary operational users of fraud-related outputs.

From an operational perspective, the component provides information that can support:

- identification of suspicious transactions
- prioritization of high-risk transactions
- passenger/ticket verification
- investigation of unusual ticket behavior; and
- decision support for railway revenue protection.

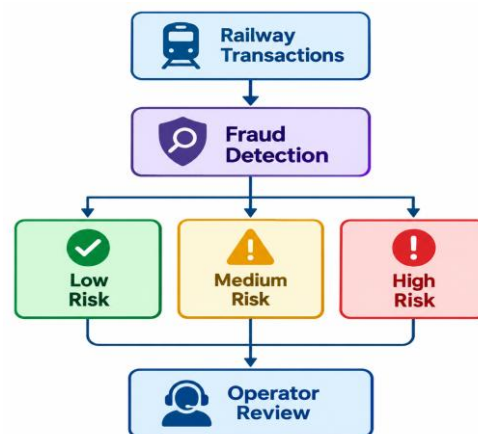
The system does not automatically determine whether a passenger is guilty of fraud. Instead, it provides risk information that can be reviewed by an authorized railway officer.

6.3.2 Operator Decision Support

The risk-based design is particularly relevant to railway operations because many transactions may need to be processed.

The system can help priorities transactions by assigning different risk levels.

For example:



This approach allows human operators to remain involved in the final decision-making process.

6.3.3 Validation Limitation

Since formal user acceptance testing was not performed, the current research cannot quantitatively claim improvements in railway staff productivity, passenger satisfaction, or operational efficiency.

Such validation would require access to railway personnel and a controlled operational environment.

Future work can therefore include:

- interviews with railway officers;
- structured usability testing;
- operator feedback;
- Passenger verification trials;
- user acceptance testing; and
- field evaluation using live railway transactions.

6.3 Integration Results

The fraud-detection component was developed as part of the wider CeylonRail Intelligence platform.

The overall platform contains four main intelligent components:

1. Passenger Demand Forecasting and Adaptive Seat Allocation;
2. Real-Time Train Tracking and Delay Prediction;
3. AI-Based Railway Fraud Detection and Passenger Verification; and
4. Multi-Objective Train Scheduling Optimization.

The integration architecture allows these components to operate through a shared platform rather than as completely independent applications.

6.4.1 Component Integration

The fraud-detection component communicates through the application architecture involving the frontend, backend services, database, and AI processing components.

Architecture allows the fraud component to use common railway information while maintaining its own specialised processing.

6.4.2 Data Integration

The fraud component can utilize transaction and passenger-related information available within the shared railway data environment.

The shared data architecture supports the exchange of information between components.

For example, passenger and ticket information can be relevant to both demand-related analysis and fraud verification, while route and train information can provide contextual information for transaction validation.

6.4.3 Backend Integration

The Flask backend provides the communication mechanism between the React frontend, database, and fraud-detection processing.

This provides separation between:

- user interface;
- application logic;
- data storage; and
- machine-learning functionality.

This modular structure makes it possible for individual intelligence components to be developed and maintained separately while remaining part of the same overall platform.

7 DISCUSSIONS

7.1 Interpretation of Findings

The research has proved that the implementation of anomaly detection through machine learning can be done for detecting frauds in the rail transaction process of Sri Lanka Railway. For this purpose, approximately 9,800 transaction records with 18 attributes were used and Isolation Forest, One-Class SVM, and Autoencoder were examined.

From the **results obtained**, it is possible to say that all three methods demonstrated different results. Isolation Forest had a ROC-AUC of approximately 0.81, while One-Class SVM and Autoencoder demonstrated a ROC-AUC of approximately 0.96. However, One-Class SVM and Autoencoder were more efficient in terms of recall. Thus, One-Class SVM and Autoencoder performed better in identifying labelled fraudulent transactions in the test dataset.

Low precision indicates the existence of false positive results. In such a way, an anomaly cannot be considered as **evidence** of fraud. The suggested system should be used as the system of decision support, where suspicious transactions should be verified by people.

Results of the risk scoring showed that among 2,750 transactions of the 2024 test set, 2,271 transactions were identified as low risk, 328 transactions as medium risk, and 151 transactions as high risk. Frauds appeared more often with **high-risk** score. It means that risk-based output can be helpful in **prioritizing** transactions for investigation.

In such a way, it was shown the technical possibility of anomaly detection for railway fraud analysis.

7.2 Comparison with Literature and Existing Solutions

Previous research has investigated railway ticket verification, blockchain-based ticketing, facial recognition, behavioral analysis, and machine-learning-based fare-evasion detection. These studies demonstrate the value of automated verification and behavioral analysis in transportation systems.

The proposed research differs by focusing on transaction-level anomaly detection and risk scoring using Sri Lankan railway data. It evaluates three different approaches—Isolation Forest, One-Class SVM, and Autoencoder—providing a comparison of tree-based, boundary-based, and neural-network-based anomaly detection.

The research also extends existing work by integrating fraud detection and passenger verification into the broader CeylonRail Intelligence platform, which combines demand forecasting, delay prediction, fraud detection, and train scheduling.

7.2 Research Contribution

The research contribution can be considered at several levels.

7.3.1 Application of Anomaly Detection to Sri Lankan Railway Transactions

The main research contribution is the investigation of anomaly-detection techniques for identifying suspicious railway transactions within the Sri Lankan railway context.

The study evaluates three different approaches:

- Isolation Forest;
- One-Class SVM
- Autoencoder

This provides an empirical comparison of different anomaly-detection approaches using the available railway transaction data.

7.3.2 Behavioral and Transactional Feature Analysis

The research considers multiple characteristics of railway transactions rather than relying on a single fraud indicator.

The feature design incorporates information related to:

- passenger behavior
- ticket usage
- route
- fare
- booking
- payment
- scanning

- inspection.

7.3.3 Risk-Based Fraud Detection

A further contribution is the use of risk scoring rather than relying exclusively on binary fraud classification.

The resulting risk levels—low, medium, and high—provide a way of prioritizing suspicious transactions.

This is particularly suitable for an operational railway environment where an operator may need to examine many transactions and cannot manually investigate every transaction with equal priority.

7.3.4 Combination of Machine Learning and Railway Rules

The research does not treat machine learning as the only source of fraud .

Machine-learning anomaly detection is complemented by railway-specific validation rules. This allows learned behavioral patterns to be considered together with explicit railway requirements.

Such a combination is useful because a transaction can be suspicious because of an unusual statistical pattern, because it violates a known railway rule, or because both conditions occur simultaneously.

7.3.5 Sri Lankan Research Context

An important contribution is the focus on the Sri Lankan railway environment.

Much of the related literature reviewed in Chapter 2 concerns general fraud detection, public transportation, or railway systems outside Sri Lanka. The present research applies these concepts to Sri Lankan railway transaction data and incorporates them into the wider CeylonRail Intelligence platform.

7.3 Product Contribution

In addition to the research contribution, the project produced a technical product component that forms part of the wider CeylonRail Intelligence system.

The individual product contribution consists of a fraud detection and passenger-verification component capable of:

1. processing railway transaction data;
2. preparing fraud-related features;
3. applying anomaly-detection models;
4. generating anomaly results;
5. generating risk scores;
6. categorizing transactions by risk;
7. supporting passenger and ticket verification;
8. communicating results through a backend service; and
9. presenting results through the application interface.

The implementation uses Python for the analytical component, React for the frontend, Flask for backend communication, and MySQL for data storage.

The component therefore extends beyond an isolated machine-learning experiment and provides an application-oriented implementation.

Integration Contribution

The product is also designed as a component of the complete CeylonRail Intelligence platform.

The wider system contains four major intelligence components:

- passenger demand forecasting and adaptive seat allocation;
- train tracking and delay prediction;
- fraud detection and passenger verification; and
- multi-objective train scheduling.

The fraud component contributes transaction-level intelligence to this integrated railway decision-support environment.

This architecture allows fraud-detection functionality to use shared railway information and communicate through the common platform rather than functioning as a separate application.

7.4 Limitations and Threats to Validity

Several limitations should be considered when interpreting the findings.

7.5.1 Dataset Size and Diversity

Although the dataset contains approximately 9,800 transactions, it represents a limited sample compared with the volume of transactions that could be generated by the complete Sri Lankan railway network.

A larger dataset containing more routes, passenger groups, ticket types, and operational conditions could produce different results.

7.5.2 Label Quality

The dataset contains labelled fraud records, and the labels were assigned according to the available transaction details. However, the reliability of machine-learning evaluation depends on the correctness and completeness of these labels.

Incorrect or incomplete labels can affect precision, recall, F1-score, and ROC-AUC.

7.5.3 Limited Real-Time Validation

The current research was conducted using available transaction data rather than a long-term live deployment within Sri Lankan railway operations.

Therefore, the results demonstrate experimental performance rather than continuous real-world operational performance.

7.5.4 False Positives

The relatively low precision values indicate that some transactions identified as suspicious were not labelled as fraudulent.

This is an important limitation because excessive false alerts could increase the workload of railway staff.

For this reason, the proposed system should support human review rather than automatically taking punitive action based solely on an anomaly score.

7.5.5 Lack of Formal Stakeholder Evaluation

Formal interviews, surveys, or large-scale usability studies involving Sri Lankan railway officers were not conducted.

Consequently, the research cannot quantitatively establish user satisfaction, operator acceptance, or improvements in staff productivity.

7.5.6 Model Generalization

The models were evaluated using the available dataset and a 2024 test period. Their performance on future transactions or significantly different railway operating conditions remains uncertain.

Changes in passenger behaviour, ticketing systems, fraud strategies, railway policies, and digital payment mechanisms could affect model performance.

7.5.7 Passenger Verification Scope

The broader system design identifies QR validation, ticket ownership validation, and identity verification as relevant passenger-verification mechanisms. However, the available transaction dataset primarily supports transaction-level fraud analysis.

Therefore, claims about biometric or advanced identity verification should not be interpreted as fully validated by the current experimental results unless those mechanisms are separately implemented and tested.

7.5.8 Integration Scope

The integrated CeylonRail architecture demonstrates how the four intelligent components can communicate through a common platform. However, full production-scale railway integration would require additional testing involving live railway systems, security controls, operational databases, and real-time data sources.

7.5.9 Threats to Validity

The main threats to validity are therefore:

- limited dataset diversity;
- possible label uncertainty;
- potential class imbalance;
- false-positive predictions;
- absence of long-term live deployment;
- absence of formal stakeholder validation; and
- possible changes in future fraud behavior.

These limitations do not invalidate the experimental findings but define the boundaries within which they should be interpreted.

7.5 Implications

The findings have implications for both railway research and potential railway-system development in Sri Lanka.

7.6.1 Implications for Railway Fraud Detection

The results indicate that machine-learning-based anomaly detection can provide an additional layer of analysis for railway ticket and transaction monitoring.

Instead of depending entirely on manual inspection or fixed rules, railway authorities could potentially use automated analysis to identify transactions requiring additional attention.

The results also suggest that different anomaly-detection approaches can produce substantially different results. Therefore, model selection should be based on empirical evaluation using railway-specific data rather than assuming that a particular algorithm will perform best in every railway environment.

7.6.2 Implications for Railway Operators

For railway operators, the risk-scoring approach provides a mechanism for prioritizing potentially suspicious transactions.

A high-risk transaction could be reviewed more urgently, while lower-risk transactions could require less immediate attention.

However, the model should remain a decision-support mechanism. Human verification remains important because an unusual transaction is not necessarily fraudulent.

7.6.3 Implications for Sri Lanka

The research has relevance to the digital transformation of Sri Lankan railway operations.

The project demonstrates how railway transaction data can be combined with machine learning and application technologies to create an intelligent railway-management capability.

The Sri Lankan context is important because solutions developed for other countries may depend on different railway ticketing systems, passenger behaviors, routes, operational rules, and data structures.

Therefore, locally relevant railway data and validation rules are important when developing an intelligent fraud-detection solution for Sri Lanka.

7.6.4 Implications for the CeylonRail Platform

At the broader system level, the fraud-detection component demonstrates how a specialized AI capability can be incorporated into an integrated railway intelligence platform.

Fraud information can potentially complement other railway intelligence functions. For example, passenger and ticket information can provide additional context for railway demand analysis, while route and operational information can support more context-aware transaction validation.

This supports the overall CeylonRail concept of sharing railway-domain information between specialised intelligent components.

7.6.5 Implications for Future Research

Future research can extend the current work by:

- collecting larger real-world Sri Lankan railway datasets;
- incorporating real-time ticketing data;
- expanding passenger-verification mechanisms;
- incorporating QR validation data;
- investigating additional anomaly-detection algorithms;
- addressing class imbalance;
- improving explainability;
- performing formal railway-operator usability studies;
- conducting long-term field validation; and
- continuously updating models as fraud patterns change.

Future research could also investigate whether combining the outputs of multiple anomaly-detection models produces more reliable risk assessments than using a single model.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

This research focused on developing an AI-Based Railway Fraud Detection and Passenger Verification component as part of the CeylonRail Intelligence platform. The study addressed the need for data-driven monitoring of potentially fraudulent and abnormal railway transactions within the Sri Lankan railway context.

The solution used approximately 9,800 records with 18 attributes and evaluated three anomaly-detection techniques: Isolation Forest, One-Class SVM, and Autoencoder. An 80/20 split was used, with 2,750 transactions from 2024 forming the test set.

The results demonstrated that machine learning can identify unusual transaction patterns. One-Class SVM and Autoencoder achieved stronger performance than Isolation Forest, with both achieving approximately 0.94 recall and 0.96 ROC-AUC. Isolation Forest achieved 0.50 recall and 0.81 ROC-AUC. However, relatively low precision values highlight the presence of false positives and the need for human verification.

A risk-based mechanism was also implemented. Of the 2,750 test transactions, 2,271 were classified as Low Risk, 328 as Medium Risk, and 151 as High Risk. This provides a practical method for prioritising suspicious transactions rather than treating all anomalies equally.

The component was implemented using Python, React, Flask, and MySQL and integrated into the wider CeylonRail platform. Overall, the research demonstrates the technical feasibility of AI-based anomaly detection and risk scoring for Sri Lankan railway transactions. However, the system remains a research prototype and requires larger datasets, real-world validation, and operational testing before production deployment.

8.2 Future Directions

Several areas can be explored to improve the proposed system.

8.4.1 Larger and More Diverse Datasets

Future research should use larger datasets covering more routes, passenger groups, ticket types, booking channels, and operational periods.

8.4.2 Improved Fraud Labelling

Verified fraud investigation outcomes could be used to improve the reliability of fraud labels and model evaluation.

8.4.3 Hybrid and Ensemble Models

Future work could combine Isolation Forest, One-Class SVM, and Autoencoder outputs to develop a more robust fraud-risk score.

8.4.4 Reducing False Positives

Feature engineering, threshold optimization, additional contextual information, and supervised learning could be investigated to improve precision while maintaining recall.

8.4.5 Real-Time Fraud Detection

Future implementations could analyze transactions during booking, purchasing, or ticket scanning and provide real-time alerts to authorized railway personnel.

8.4.6 Enhanced Passenger Verification

QR validation, stronger digital ticket verification, and authorized identity-verification mechanisms could be incorporated where appropriate and legally permitted.

8.4.7 Live Railway Integration

Future work could integrate the system with real ticketing, inspection, scheduling, and passenger-information systems.

8.4.8 Continuous Model Monitoring

Model performance could be continuously monitored and models retrained as passenger behavior and fraud patterns change.

8.4.9 Stakeholder Evaluation

Future studies should involve railway staff and other relevant stakeholders to evaluate usability, interpretability, and operational suitability.

8.4.10 Production Deployment and Security

Future deployment should address authentication, authorization, secure APIs, database security, audit logging, privacy, backups, monitoring, and system availability.

8.4.11 Expansion of Railway Intelligence

Fraud patterns could eventually be analysed alongside other CeylonRail components to identify trends by route, time, booking channel, and passenger activity, supporting broader railway decision-making while maintaining appropriate privacy controls.

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GLOSSARY

Term	Definition
Artificial Intelligence (AI)	A field of computing concerned with developing systems that can perform tasks that normally require human intelligence, such as recognizing patterns, analyzing information, and making predictions.
Anomaly	A transaction or behavior pattern that differs significantly from the expected or normal behavior in a dataset.
Anomaly Detection	A machine learning approach used to identify observations that appear unusual compared with most of the data.
Autoencoder	A neural network model that teaches to reconstruct input data. In anomaly detection, unusually high reconstruction errors can indicate an abnormal transaction.
Booking Channel	The method through which a railway ticket or transaction is created, such as an online or other supported booking method.
CeylonRail Intelligence	The broader intelligent railway platform developed as the group project, consisting of multiple AI-supported railway management components.
False Positive	A legitimate transaction incorrectly identified by the system as suspicious or fraudulent.
F1-Score	An evaluation metric that combines precision and recall into a single measure using their harmonic mean.
Fraud Detection	The process of identifying transactions or activities that may represent fraudulent or unauthorized behavior.
Fraud Risk Score	A numerical or categorized indication representing the level of risk associated with a railway transaction.
Isolation Forest	An unsupervised anomaly detection algorithm that identifies unusual observations by isolating them from other observations through random partitioning.
Machine Learning	A branch of AI in which computational models learn patterns from data and use those patterns to make predictions or identify unusual behavior.
MySQL	A NoSQL database used in the developed system for storing and managing application-related data.
One-Class SVM	A machine learning algorithm designed primarily for identifying observations that differ from a learned representation of normal data.

Passenger Verification	The process of checking whether passenger and ticket-related information is consistent and valid according to the available railway transaction information and validation rules.
Precision	The proportion of transactions identified as positive or suspicious by a model that are actually positive according to the evaluation labels.
Recall	The proportion of actual positive or fraudulent transactions that are successfully identified by the model.
Risk Category	A classification used to communicate the estimated level of concern associated with a transaction, such as Low, Medium, or High.
Risk Scoring	The process of assigning a numerical value or category to represent the potential risk associated with a transaction.
ROC-AUC	A performance measure based on the Receiver Operating Characteristic curve that evaluates how well a model distinguishes between different classes across classification thresholds.
Route Inconsistency	A situation where the route-related information associated with a transaction differs from an expected or valid travel pattern.
Suspicious Transaction	A transaction containing characteristics or patterns that cause the system to flag it for further examination.
Ticket Scan Count	The number of recorded scans associated with a railway ticket or transaction.

APPENDIX

Appendix A UI interfaces of Advanced Intelligent Ticket Fraud Detection and Risk-Aware Passenger Verification

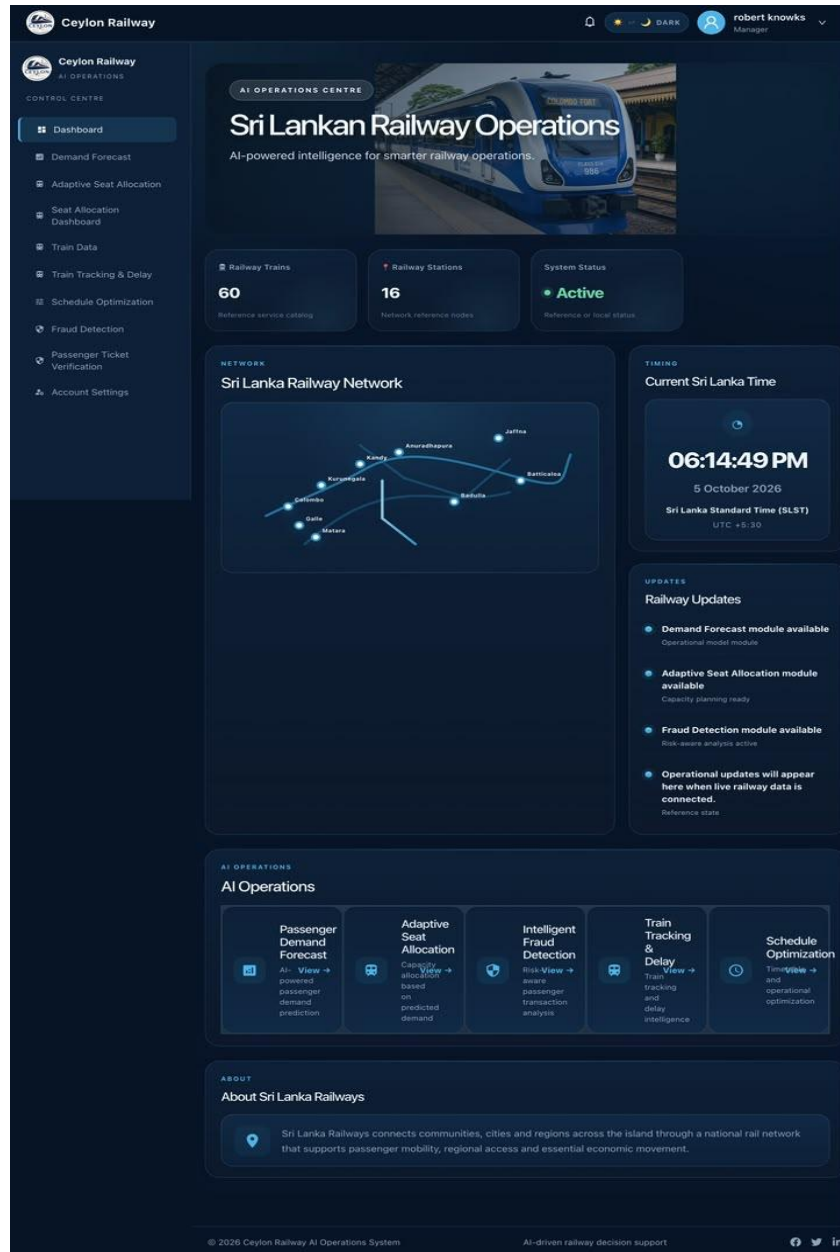


Figure 16 - User Interface 01

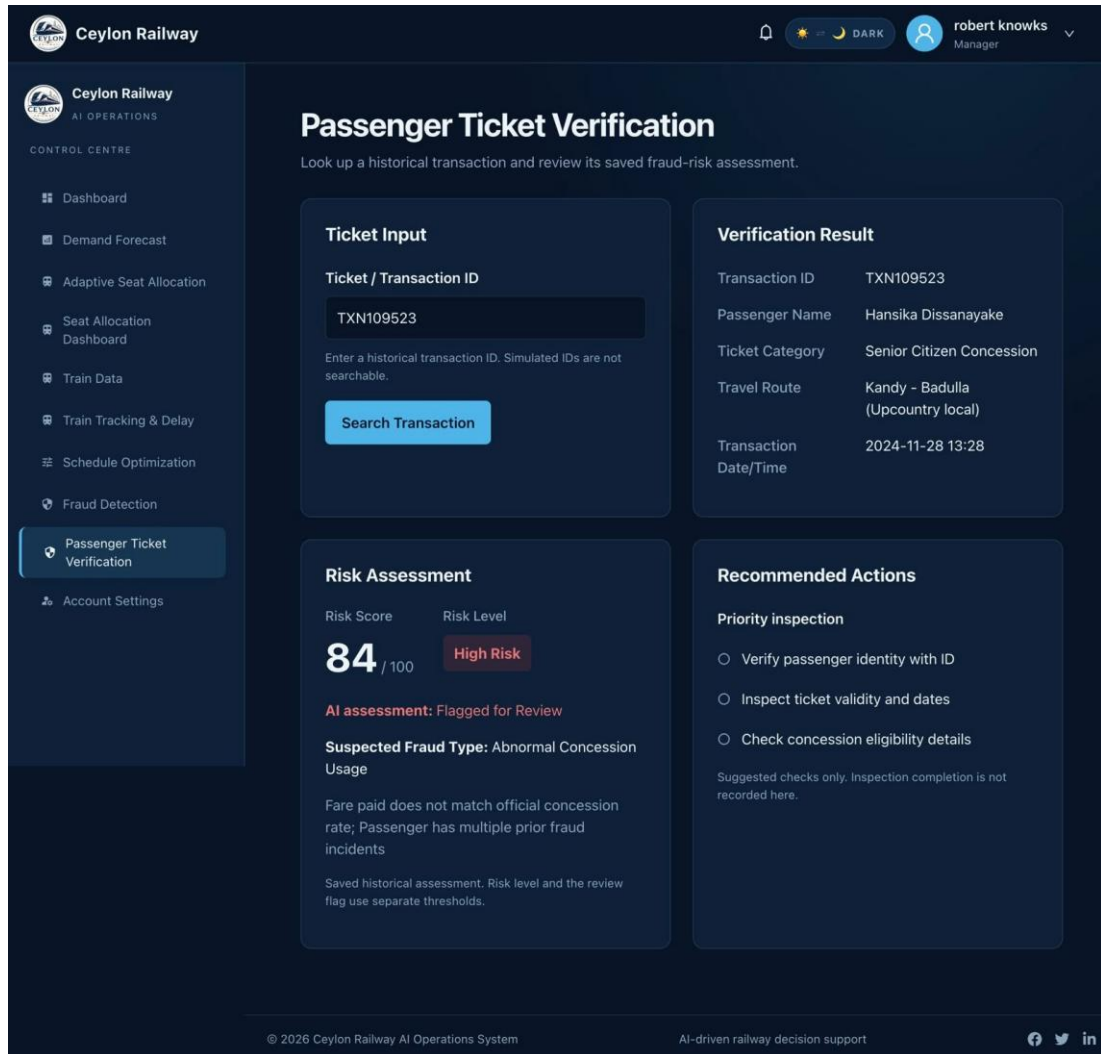


Figure 17 - User Interface 02

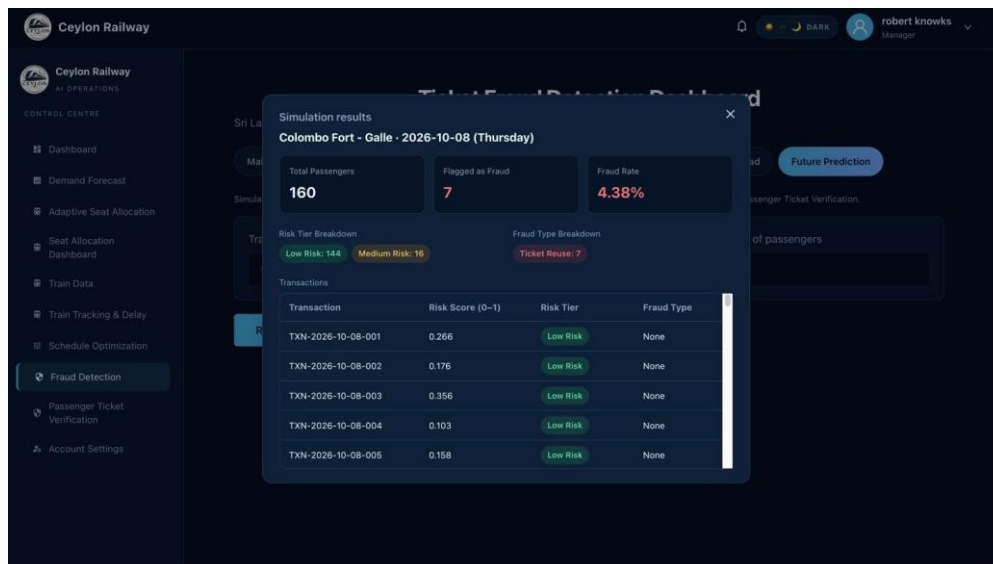


Figure 18 - User Interface 03

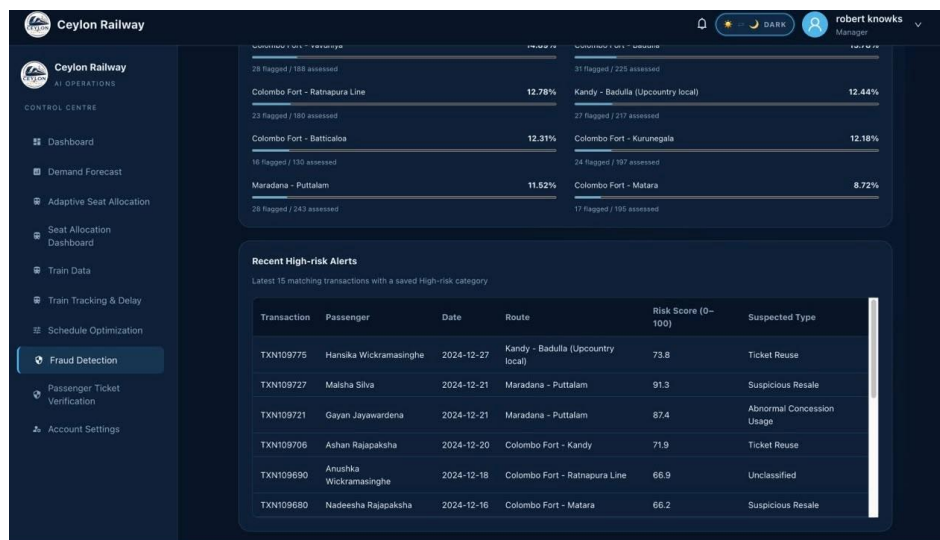


Figure 19 - User Interface 04

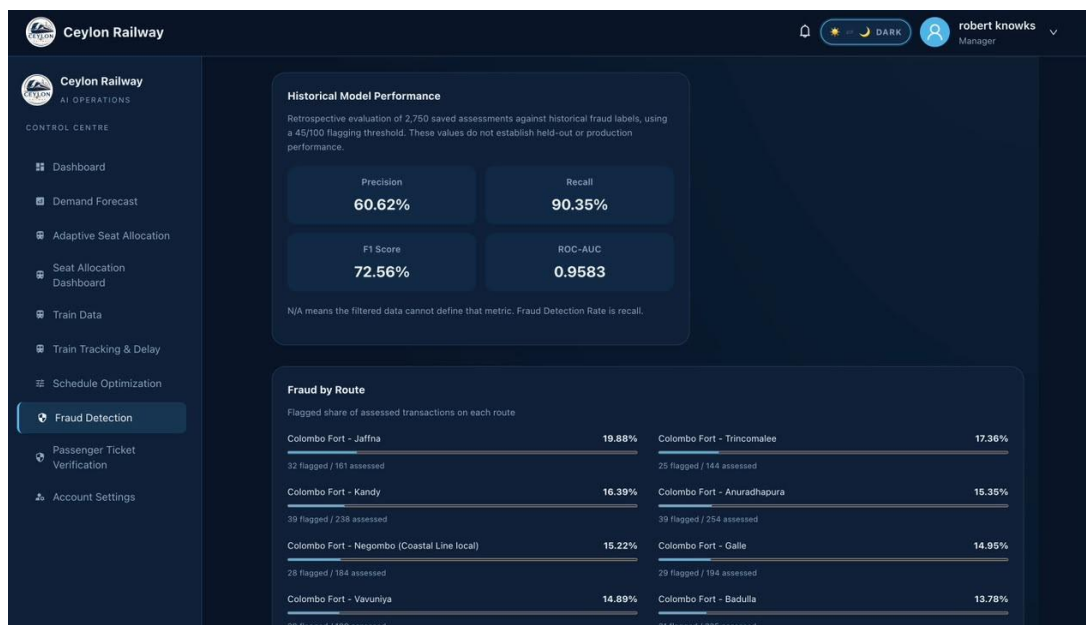


Figure 20 - User Interface 05



Figure 21 - User Interface 06